Discoveries

Original findings from a grown spiking creature, with no external antecedent

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The `aibaby` project — a C++17 spiking neural creature grown from a DNA/genome file

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Abstract

This document collects the findings of the `aibaby` project that are not held up by any published result. The project grows a spiking neural "creature" from a genome file and teaches it to produce vowel formants; much of its machinery is named after the literature it came from, and those mechanisms are deliberately excluded here. What remains is the set of things the system itself was the first to say: measured quantities with no external reference, structural obstacles that only appear in this architecture, and a group of inference failures that were found by being committed and then caught. The longest single thread is §4: because the readout is a rate-weighted centroid, the influence of one neuron on behaviour can be differentiated rather than estimated, and that derivative — taken with a sign, which is the part the data forced — accounts for a measured asymmetry, its reversal between targets, and why one restriction on plasticity costs half the lesson while another binds a third of all synapses and costs nothing.

The methodological findings are listed first and at length, because they are the most transferable part and the least specific to a creature made of spikes. Several of them are reports of a mistake that survived review — a test no outcome could pass, a refusal criterion computed from the wrong arm, a statistic diagnosed in a comment and then quoted for months. They are included on the view that a failure mode with a worked example is more useful than a rule without one.

Every quantity below comes from a run gated on a pinned determinism hash, `ad96f882becbee92`, under which every mechanism that ships off is bit-identical. Section 8 gives provenance.

1. Scope, and what is excluded

A finding is included if the project discovered it, and excluded if a published result says it first or frames it. The line is drawn at the *claim*, not at the tools: the estimators used throughout are Cochran's ratio estimator and the Quenouille–Tukey jackknife, and the quantities they produce are still this project's, so they appear here while the methods do not.

Excluded on that basis are: the account of interference as failed contextual inference (Heald, Lengyel and Wolpert's COIN model), the finding that context gating alone is insufficient without synaptic stabilisation (Masse, Grant and Freedman), importance-weighted protection of earlier weights (Kirkpatrick et al.; Zenke, Poole and Ganguli), node perturbation as a motor learning rule (Fiete, Fee and Seung), the stability–plasticity dilemma (Grossberg), the uniform cluster prior implicit in MacQueen's `1/wins` rule, the division of a population code into output-potent and output-null dimensions (Kaufman, Churchland, Ryu and

Shenoy), and activity-dependent myelination as a consolidation mechanism (the oligodendroglia literature, cited by title and DOI in the main list where author lists could not be verified). Each is cited in the project's main reference list. Appendix A lists the project results that were excluded by this criterion, so the exclusion can be audited rather than taken on trust.

2. Measurement and inference

These came out of running the same experiment many times and being wrong in a way that printed a number. None has an external source; each has a worked example in the project history.

2.1 Chance has no fixed value when the readout is "nearest"

A `k`-alternative naming score read against `1/k` is not a chance comparison. When the decision rule is nearest-target in formant space, chance depends on the *geometry* of the targets: the same uninformative voice scores 0.161 against targets on the vowel corners and 0.306 against a 2×2 arrangement. An informationless voice reads about 0.30 on a four-word task where `1/k` would predict 0.25. One earlier ceiling claim was a null error of exactly this kind: a real effect scored against an arithmetic chance level while the matched control sat in the same table, unread.

2.2 A test that nearly all outcomes pass is not a test, and neither is one that none can pass

Two symmetric failures, both found after they had produced a published-looking verdict. A guard written as `cd2 > −3·13.1` — "greater than −39.3" — passes on nearly any data, and was reported as "F2 is spared" when it meant "F2 is unmeasured". In the other direction, an intervention's success branch required a shift of 0.090 where a complete closure of the gap could produce only 0.070; it printed a confident conclusion from an unreachable branch. The fix in both cases is to print the power of the test beside its result, so that a criterion nothing could have cleared is visible rather than silent.

A third shape, and the one that hid longest: a criterion only the best possible result fails. An intervention's success branch asked whether the residual effect was indistinguishable from zero, written as $|\text{residual}| < 2 \times$ its standard error. The intervention then worked *perfectly* — the residual came back exactly zero, so its standard error was exactly zero too, and $0 < 0$ is false. Execution fell through to the weaker branch and printed "contributes but does not account for it" over numbers showing a complete closure at 5.5 SE. The two earlier instances were a criterion nothing could clear and one almost anything could, which is why neither suggested looking for this: here the test is passed by every outcome *except* the strongest. The general fix is to state a closure as a fraction of the quantity being closed rather than as a distance from zero, because the distance-from-zero form divides by a precision that a perfect result drives to nothing.

2.3 The reference point is the hypothesis's own prediction, not zero

A hypothesis that names a value must be tested against that value. A drift account predicting a decay of −0.034 was first tested for being distinguishable from *zero*, which it was not, and so survived; tested against its own prediction, the measurement sat 3.9 SE away and the account was excluded. The same error in another guise is scoring an effect against `1/k` while the real control is in the table.

2.4 A null and a reversal are different results

An `else` branch that catches both reports the wrong one. An intervention expected to close a gap was tested only for closure; when the gap *widened*, the outcome fell through to the null branch and was written up as "no effect". The widening was 2.5 SE in the opposite direction to the prediction, which is a stronger result than the null it was reported as.

2.5 A constant copied instead of derived fails silently, and travels

Six instances in one 34,000-line file, in two families. *Indices:* a literal array size that was a copy of a named bound and overran into the next field; positional grid indices; positional arm indices that would have mislabelled whole rows; and pair indices `(k·2, k·2+1)` that silently shifted when two table rows were inserted above them. *Labels:* a band value copied beside the table it came from, so an experiment reported its widest condition as a tenth of its real width after the table was rescaled; and a deleted control whose label and refusal criterion were inherited by unrelated rows; and a single hardcoded baseline arm, correct while every arm shared one lesson target and silently wrong the moment a second target was added, which charged the new arms the whole difference between the two conditions — one cost read 11% against the old control and 3% against its own, and three of eight printed numbers were wrong.

What makes this class survive review is specific and worth stating: **every site carried a comment acknowledging the indexing was positional, and the comment was treated as the fix.** One even warned about a *skipped* row reading zero — a different hazard from a *shifted* row, against which the guard written for the first does nothing. A mislabelled number still prints with a standard error and still reads as a result, so nothing in the output looks wrong. The question that finds these is: *is this label attached to the data, or merely next to it?*

2.6 A passing determinism hash does not say the code is correct

The project gates every run on a pinned hash. A pass says the shipped configuration is unchanged and the run is reproducible; it says nothing about whether the experiment computes what its name claims. Four of the six errors in §2.5 were introduced and committed under a passing hash.

2.7 Reproduce the conditions, not the experiment name

A headline was called irreproducible after three runs at the registered default tick count. The recorded result had been taken at 3.5× that budget, which was written in its own log's filename and in the note that cited it. A registered default is not evidence about what a recorded run used, and this system has a documented power curve across exactly that range — one experiment reads +1.0 at 560k ticks against +18.3 at 3.4M.

The same error in its archive form, and it is the easier one to commit. A ladder of mask extents was assembled by collecting arms from every log in the results directory with a single `grep -h` matching on the arm's *name*. One of the seven rows came from a pilot run whose own baseline arm gained +0.0363 where every other row's baseline gained +0.3618 — a tenfold difference in the quantity being compared — and nothing in the pooled output recorded which log each line came from. **A search across runs is a join on the arm name, and the arm name does not carry the budget.** The rule that catches it is to read the anchor of the log a number comes from before reading the number, which this project applies to its own runs as a

matter of course and had not applied to its own archive.

2.8 Diagnosing a broken statistic does not retire it

A retention measure was computed as a mean of per-creature ratios whose denominator — how much that creature learned — was floored only at `1e-6`. The code carried a comment beside it recording that the statistic had read −5.587 ± 6.729 and 27.222 ± 21.160 on another genome and calling it “a broken statistic reporting nothing”, and a denominator-free replacement was added alongside. The headline, the milestone gate and the pre-registered specificity criterion all went on reading the broken one.

Measured on the same 128 sessions, the two estimators disagree in *sign*: a mean of ratios gives +0.182 ± 0.663 and a jackknifed ratio of sums gives −0.211 ± 0.074, with standard errors differing ninefold. The transferable part is not the statistic but the order of operations: a comment records a fact, it does not change what the gates compute, and every consumer has to be moved by hand. Grep for the variable, not for the comment.

2.9 Unanimous signs on three seeds are not power

Repeatedly, effects that were unanimous and large on a handful of creatures did not survive more of them. A smoothing sweep read +0.24 at n = 3 and +0.03 at n = 6; a salience-reward effect was +2.6 SE at n = 12 and null at n = 16; a vigilance score of 0.737 fell to 0.594 at 96 creatures. The project adopted a 3 SE bar at $n \le 18$ as a direct consequence, and the bar must be fixed once and named, not chosen per test: one verdict printed “not yet a finding” for 2.9 SE against a 3 SE bar and then made a 2.7 SE contrast its headline because that test had been coded against 2.0.

2.10 Two points agreeing to a tenth of a percent is one coincidence

Two lesson targets gave lever costs of 30.01% and 22.12% of the lesson, and the distance each had to move the readout stood in a ratio of 1.356 against the costs’ 1.357 — a match to 0.1%, from no fitted parameter. It was recorded as a coincidence on two points rather than a law, and the third target refused it: the law predicted 14.8% and the measurement was 25.0% ± 5.9%, and the three points are *non-monotone* in distance travelled (0.200 → 25.0%, 0.300 → 22.1%, 0.407 → 30.0%), which no proportionality can produce at all.

The transferable part is the arithmetic rather than the caution. Any smooth one-parameter family can be made to pass through two points, so a two-point agreement carries no information about the family; and the cheapest way to settle it was to measure a third point, which cost one overnight run, rather than to argue about the first two. The same project had already paid for the opposite choice once: a leverage law fitted to three costs with one free parameter matched them to 1.68 percentage points and then failed two further costs out of sample. Restricted to the points where the lever is the same neuron, the surviving pattern needs a large distance-independent term plus a weak distance term — two parameters on two points — so the account was stopped there instead.

2.11 A pre-registered prediction can be wrong in sign on both of its axes

Pre-registration fixes what a result will be compared against; it does not make the prediction right, and the failure mode worth recording is the one where a prediction fails in the direction nobody wrote a branch for.

A cross-mechanism regularity — every plasticity restriction tried here cut learning more than it cut forgetting — was used to predict that removing a restriction would buy more learning than it cost in retention. The arm that removed it gained *less* and retained *less*. Both differences are nulls, so the regularity is not refuted at significance, but neither was it confirmed, and the registered text had no reading for "worse on both axes". Registering the two directions a prediction could fail in is not enough when the mechanism can also simply fail to matter.

A second shape of the same failure, from a registration that offered two accounts and asked which one the data would pick. One said a correctly aimed quarter-sized mask would keep most of the learning of a half-sized one; the other said both quarters would cost more than the half and that aiming would buy nothing. The measurement gave *both* effects, separately resolved and of comparable size: aim was worth $+0.0956 \pm 0.0063$ at matched size, and shrinking the mask at matched aim cost -0.0719 ± 0.0132. **A dichotomy cannot be confirmed by a conjunction** — it can only be half-refused on each side, which is what the write-up had to say. The discipline that would have survived is to register the quantities and the direction each should move, not a menu of accounts to choose between.

2.12 Grep the results directory before building the experiment

Three times in one week the answer to a question this project was about to spend hours measuring was already in its own output. A symmetric account of leverage was refuted by a run on disk, not by a new one. A law relating cost to distance was refused by measuring a third point rather than by arguing about the first two. And a plan to re-measure a cost curve at the opposite end of the motor group — written up, committed, costed at about five hours — turned out to be an experiment that had run three weeks earlier, whose summary was filed in the note immediately adjacent to the one being corrected, and which the same day's work had cited for a different number.

The last of the three is the instructive one, because the obstacle was not that the result was hard to find. It was that the question had been *framed as a lead*, and a lead reads as something not yet done. The habit that catches it is mechanical rather than attentive: before building an arm, search the existing logs for the condition it would create — the position, the count, the target — and not for the name of the experiment that would create it. Arms are cheap to name and expensive to run, and the name is the one part that does not persist.

The same habit has a second form, which turned up within the hour of writing the first. Searching the archive *by arm name* finds every run of that arm and silently pools runs made under different conditions — §2.7 records what that cost. Both failures are one mistake: **treating a name as an identifier for a condition.** A name identifies an intention. The condition lives in the log beside it, and has to be read from there.

2.13 A score with a margin constraint has a ceiling, and comparing raw scores across it merges two effects

The clearest instance of §2.1's error, and the one that changed a result. An unsupervised partition is scored by best-permutation agreement with the true labels — the diagonal of the label×cluster confusion matrix, maximised over assignments of clusters to labels. Reported alongside it is the share of items the busiest cluster claimed. **The two are not independent.** For any assignment the diagonal is bounded term by term by

the smaller of its row and column mass, so with label frequencies w_c and cluster shares q_c the agreement cannot exceed $\Sigma_c \min(w_c, q_{\sigma(c)})$, and for two balanced labels with busiest share q this is exactly:

$$\text{agreement} \le 1.5 - q$$

Checked on 216 creatures: no violations. It is not a vacuous bound either — a supervised control attains it exactly, and two of six arms sit within 0.004 of it. So the score splits into **a ceiling set by how lopsided the partition is, and a slack against that ceiling which is the only part that is classification error at all.**

The project had reported a difference between two conditions, reproducing on 18 of 18 creatures at 6.0 SE, as one effect across three pairs of arms. Decomposed, it is not the same effect in any two of them: in one pair the two ceilings are within 0.008 and the whole difference is classification; in another the ceilings differ by 0.055 and the classification difference is 0.001. **One of the three pairs contains no classification difference whatsoever**, and its number had been read as though it did. Two arms were also being compared as better and worse at the same task when, measured against their own ceilings, their errors were 0.004 and 0.187 — opposite ends of the scale the raw scores put next to each other.

The transferable rule is narrow and mechanical: **when a statistic is a maximum subject to margins that vary between conditions, compare the slack, not the statistic.** And the general shape is §2.1, §2.3 and §2.7 again — a reference point that exists but is not the one the claim is about. This is its fourth distinct instance here, which is why it is recorded as a class rather than as four separate mistakes.

2.14 An absence in the source is not an absence in the behaviour

Caught before it was published, and worth recording because the reasoning looked sound all the way to the conclusion. A mechanism was proposed that would only fire while the creature was asleep, so the first question was whether the relevant protocol sleeps at all — a fair question, since this project has already lost a result to a mechanism that could not fire in the window it was tested in. The protocol's source contains no mention of sleep, of waking, or of the replay that accompanies it. On that basis the lead could have been closed for nothing.

It sleeps. Sleep here is driven by an internal fatigue variable crossing a threshold, so the host never calls for it and never needs to name it. **The absence of the word meant "this code does not command it", not "it does not happen"**, and the two readings are indistinguishable from a search. What separates them is reading the mechanism that would produce the thing rather than searching for the name of it.

The general form is the one error this document keeps finding in new costumes: a reference point that exists but is not the one the claim is about. Here the claim was about a behaviour and the evidence was about a vocabulary. The habit that catches it is to ask what would have to be true in the *kernel* for the answer to be no, and then look there — which in this case took one grep and produced the opposite answer.

The lead did die, but on arithmetic rather than on absence, and the arithmetic is the interesting part. The sleep bout is fixed by two genome constants and works out at about three minutes against a roughly twenty-minute cycle, so the mechanism being proposed would run about fifteen percent of the time — and the mechanism it was to be grafted onto already runs continuously and costs nothing on either axis it was built to serve. Running a null fifteen percent as often is not a repair. That is a better refusal than the one the search would have given, and it also said what to build instead.

2.15 A guard that the mechanism is live is not a guard that the measurement can resolve it

The sharpest version of §2.2 yet, because the guards in question were written deliberately, in advance, and passed. An experiment asked whether a reward could be aimed at one phase of a self-generated two-phase movement. It carried two vacuity guards, both chosen before the run: one confirming the movement was happening, and one confirming the two phases were *balanced* rather than one having won outright. Both passed cleanly — the phase was reported on essentially every rewarded moment, and the split was 0.50.

Neither guard asked whether the **reward** could tell the two phases apart, which was the entire question. Credit is delivered through an eligibility trace, and the trace's window is 2000 ms against a phase lasting about 136 ms — roughly fifteen phases per window, seven full cycles of the movement. Both halves sit inside every credit window, equally eligible, every time. So the two conditions being compared were *the same condition with the labels permuted*, and the null that came back measured the operating point rather than the thing the experiment was named after.

The distinction worth carrying: a vacuity guard normally asks whether the treatment happened. This failure needs a guard asking whether the instrument can see the treatment. Those are different questions and the first does not imply the second. The arithmetic that settles it was available before the run — three constants, all already measured — and cost nothing but the thought of comparing them.

Recorded with its chronology because that is the only part that makes it useful rather than embarrassing. The arithmetic was done and the prediction written down while the run was still executing: that the contrast would come back null, that a resolution guard would have voided it, and that a positive result would refute the whole argument. The guard was added before the log was opened, and run afterwards on the very configuration that produced the result it prints a refusal. A verdict that is wrong over correct numbers is recoverable; what makes it recoverable is having written the correction down before the numbers arrived.

2.16 A guard that the measurement can resolve the treatment is not a guard that the treatment did anything — and the positive control must be checked against the project's own known negatives

§2.15 added a resolution guard to an experiment that had liveness guards and needed one more. The next run of that experiment carried all three — the movement is happening, the two phases are balanced, the credit window is short enough to tell them apart — plus a fourth added with them, that the credit can still *arrive* after the delay it has to survive. All four passed. The verdict printed a refusal. The refusal was not available.

What exposed it was not a guard but the results table. The measured quantity came back *identical to a previous run of the same experiment, to within 0.5%*, after the two quantities the run existed to vary had been changed twenty-fold and ten-fold respectively. A measured value that ignores both of its own causes is not measuring them. The null was equally consistent with the lesson having no effect at all, and nothing in four guards addressed that, because each one asked whether some *part* of the apparatus was working rather than whether the *intervention* had an outcome.

So the ladder has four rungs and they are all different questions. Is the mechanism present? Is it live? Can the instrument resolve it? And did the treatment produce any effect at all on anything? A positive control answers the fourth and only the fourth, and the first three do not imply it. Both halves of the quantity that answers it — error in the first third of the session and in the last third — had been recorded by the kernel

all along and neither had ever been printed. Two runs reported *where* a lesson had landed without ever showing that one arrived.

And the control is easy to get wrong in two further ways, both of which were caught only while building it. First, it cannot be read on a treated arm: the error there is scored on a trial-mean output against a target that changes within the trial, so no single output can satisfy it and the control would have voided a working experiment. Second, and worse, the obvious candidate was a configuration this project had *already measured as a null* — a lesson aimed at a heard target, recorded years-equivalent earlier at −0.1% against +24% for a fixed one. A positive control that is already known not to work establishes nothing, and it would have voided every run of the experiment for a reason unconnected to the question. **A positive control is a claim about the apparatus, so it has to be chosen against the record of what the apparatus is known to do**, not against what it ought to do.

The control, once built correctly, passed at 6.3–18.4 SE on three genomes, so the refusal it was built to check survived. That is the outcome worth noticing: the guard did not overturn the result, it made the result defensible. The version without it was the same number with no entitlement to it.

2.17 A sharper statistic applied to the sample that produced a claim lands it on the bar

A lead stood at **+0.0256 ± 0.0117 (2.2 SE)** on 18 creatures: narrowing an eligibility window twentyfold appeared to *improve* a fixed-target lesson while delivering strictly less credit. Picking it up revealed that it had been measured loosely. The two genomes compared differ in exactly one scalar, which does not re-roll the wiring, so each seed names the same creature in both runs and the comparison was **paired by construction** — it had only ever been quoted unpaired because the per-creature quantity was never printed, existing solely as a mean. Pairing was therefore available for free, and the obvious move was to take it.

sample	paired difference		sign count
all 36 creatures	**+0.0154 ± 0.0080**	1.9 SE	26 of 36
the original 18	**+0.0256 ± 0.0084**	**3.0 SE**	15 of 18
18 the claim had never seen	**+0.0052 ± 0.0135**	0.4 SE	11 of 18

Pairing reproduced the original point estimate to four decimals and carried it to exactly 3.0 SE — onto this project's bar — on the creatures that had produced it, while 18 fresh creatures give a fifth of the effect and nothing resolvable. Had the existing data merely been re-analysed, which costs nothing and was the more obvious of the two moves, the result would have been published at the bar and would have been noise. The 3 SE threshold is a statement about sampling error in data the claim has not seen; it carries no protection when the claim and the sample share an origin, and a sharper instrument aimed at the discovery sample simply re-finds the discovery.

A second, narrower finding sits inside the same run and corrects an expectation stated before it: **pairing bought almost nothing here**, moving the pooled standard error only from 0.0086 to 0.0080, because the across-creature correlation between the two conditions is **r = +0.133**. A paired design converts between-unit variance into precision, so it pays in proportion to how strongly the units correlate, and here the measured quantity is dominated by within-unit noise rather than by which unit it is. What settled this question was

doubling the creatures, not sharpening the statistic. The same structure of fact refused a two-pool readout elsewhere in this project, where only $\rho = 0.101$ of the relevant variance proved to be shared.

The lead itself is *refused* at 1.9 SE, and the distinction matters: this is a failure to replicate, not a demonstration of absence. The fresh half's interval extends to +0.032 and so cannot exclude the original effect — only fail to find it. The effect size fell 40% as the sample doubled, which is the same signature that retired an earlier reward manipulation in this project when it shrank between $n = 12$ and $n = 16$.

2.18 A decoder's accuracy is a lower bound, so the gap between two decoders measures the decoders

A run was built to decide whether a two-bit condition code was limited by what the ear carries or by how the index is computed, and its pre-registered primary had to be retired. The design compared, for each bit, how well a supervised readout recovered it against how well the clustering rule did — and read a 5.4-standard-error asymmetry between the two bits.

The statistic cannot bear that reading. Every column in such a table is a decoder scored on held-out data, so every column is a *lower bound* on what the signal holds about its label, never an estimate of it. Two bounds cannot be subtracted: each decoder is lossy, and how lossy depends on the problem, so a reader that compresses a hard quantity more than an easy one manufactures an asymmetry from a symmetric signal.

The table refuted its own primary from inside, which is what makes this demonstrable rather than arguable. Applying the same supervised reader to a normalised version of the identical feature dropped one bit from 0.850 to 0.646 — 0.204 of content lost to the reader alone — and the apparent asymmetry *grew* to 9.4 standard errors. An asymmetry that moves with the reader is a fact about readers.

What such a run can report is bounds: here the feature holds the two bits at at least 0.916 and at least 0.965, and those best bounds differ by 0.049 where the reader-based gaps read 0.115 and 0.305. **To compare content across two quantities one needs a ceiling, not another decoder** — an oracle, or an exact decomposition with no free parameters. The within-column comparison that holds one quantity fixed and varies the *rule* remains sound, and that is where the same run's usable result came from.

3. Credit assignment and interference

Teaching a second, conflicting vowel target destroys the first. The surviving fraction is 0.22. This section is what the project established about why, and about what does not fix it.

3.1 The wipe has a mechanism, and it is not a representational collision

Two lessons interfere while driving disjoint neuron groups, so the groups are not what they share. What they share is the reward: a single broadcast scalar nudges the bias of every neuron in the module, so teaching B random-walks A's setting. Measured directly, B restores the very neurons A suppressed. Orthogonal targets are held simultaneously at 0.84 against the conflicting pair's 0.22, which locates the failure in the conflict rather than in capacity.

3.2 It is an axis collision, not a distance effect

A second lesson that is silent about the first lesson's axis is free; one that speaks to the same axis is

destructive, confirmed at +11.5 SE. Distance between the two targets is not the variable.

3.3 A perfect per-neuron credit mask does not remove the wipe

An oracle mask — the host tells the creature which lesson is live, and the reward is confined to the half of the motor group that lesson uses — retains *less* than the broadcast rule it was meant to repair: +0.109 ± 0.057 against +0.320 ± 0.052, a deficit of −0.211 ± 0.074, negative at every gain floor. It also learns less, gaining +0.196 ± 0.015 against +0.362 ± 0.033. This retracts an earlier claim of the opposite sign, which had rested on the estimator described in §2.8.

3.4 The mask's cost decomposes additively, and content and position cost the same

On end-of-teaching error, paired over 32 creatures:

Step	Contrast	Effect
masking at all	no mask → mask aimed correctly	+0.211
content	correct aim → random aim	**+0.103**
position	random half → central half	**+0.102**
total	central mask vs no mask	+0.417 ± 0.044 (9.5 SE)

The two increments are equal to within 0.001, and each is about half the base cost of masking at all. The direct position contrast — a central mask against a randomly aimed one, at matched content and matched mask fraction — is +0.102 ± 0.009, 11.5 SE.

The ordering on learning is: no mask, then correct aim, then random aim, then central. That a mask excluding *both* extremes of the group costs as much again as mis-aiming one is consistent with the leverage geometry of §4.

3.5 The aim of the mask during the interfering lesson recovers half of what masking costs

Two arms that mask the first lesson identically and differ only in which half they confine the reward to during the *second* lesson. Both therefore learn the first lesson equally well — verified by end-of-teaching error identical to +0.0000 ± 0.0000 over 32 creatures — so the comparison is clean.

Gain floor	Aim correct	Aim wrong	Difference	n
0.00	+0.1092	−0.0873	−0.1965 ± 0.0510 (3.9 SE)	32
0.05	+0.1291	−0.0815	−0.2107 ± 0.0509 (4.1 SE)	29
0.10	+0.1327	−0.0782	−0.2109 ± 0.0508 (4.2 SE)	27
0.20	+0.0598	−0.0934	−0.1532 ± 0.0478 (3.2 SE)	18

Retention as a paired, jackknifed ratio of sums, swept over a floor on per-creature learning. The effect holds at every floor that retains enough creatures, which is the criterion the experiment's own verdict gate applies.

Against a no-mask baseline of +0.320, a wrongly aimed mask sits −0.407 below it and correct aim recovers +0.197 — about 48%. The residual −0.211 is the part no amount of correct aim removes. So credit

assignment here is not inert: what defeats the oracle is that the price of confining the reward exceeds what aiming it well buys back.

3.6 Position and content are not separable in a binary half-split

A structural result, and a negative one about the design rather than the creature. The mask is a contiguous half of the motor group and each lesson prefers exactly one half, so within a phase "the mask is in the wrong place" and "the mask selects the wrong neurons" are the same statement. No contrast built from end-aligned masks can separate them.

The only mask aligned with neither lesson — the central half, which leaves each lesson half of its preferred neurons on every trial — does separate them, and cannot answer the retention question for a second structural reason: it cuts learning by a factor of 29 (gain +0.0125 ± 0.0023 against +0.362), so no creature clears even the lowest meaningful gain floor and retention has no denominator. **Varying position far enough to matter destroys the learning whose retention one would be measuring.** More creatures cannot fix this; it is not a power problem.

3.7 No mask in this family is cheap enough, and that closes the line by arithmetic

The two numbers above are a sum: confining the reward costs −0.211 ± 0.074 of retention and aiming it well recovers +0.197, so credit assignment here is outbid rather than inert. A cheaper mask was therefore the only thing that could change the sign, and §4.2 said where to look — most of the oracle's half is low-potency ballast, so a small mask flush against the lever should keep the learning and pay less for it. Measured across every mask extent at the best available aim, on one genome, 32 creatures each:

Reward reaches	Aimed at	Learning gained
14 of 14 (no mask)	—	**+0.3618 ± 0.0334**
7 of 14	the lever end	+0.1964 ± 0.0149
4 of 14	the lever end	+0.1245 ± 0.0046
7 of 14	a coin flip each trial	+0.1022 ± 0.0076
4 of 14	the opposite end	+0.0289 ± 0.0054
7 of 14, centred	neither end	+0.0125 ± 0.0023

Aim at matched extent is +0.0956 ± 0.0063 (15.2 SE) paired on seed; extent at matched aim is −0.0719 ± 0.0132 (5.4 SE).

Completing that as a square — extent against which end — gives the sharper form, with both wrong-end cells measured at the same budget:

	reward reaches 7 of 14	reward reaches 4 of 14
the lever end	+0.1964 ± 0.0149	+0.1245 ± 0.0046
the opposite end	+0.0265 ± 0.0041	+0.0289 ± 0.0054

Going from seven to four costs −0.0719 ± 0.0132 (5.4 SE) at the lever end and +0.0024 ± 0.0066 (0.4 SE) at the other. The

second contrast's standard error is small enough that a step the size of the first would have read 10.8 SE.

Extent is a second axis and potency gates it. An account in which extent is simply signed potency summed over the retained neurons predicts that restricting *fewer* wrong-end neurons does *less* damage; the half and the quarter at the wrong end are instead indistinguishable, at power to have seen a tenth of that. Where there is no potency, extent does nothing — not a reversal, nothing. §4.5 reaches the same square in the freezing family, and had predicted this cell before it was measured.

Learning rises monotonically with how much of the group the reward can reach, and the cheapest mask is no mask. The best-aimed quarter keeps 63% of the half's learning on 57% of its neurons, so potency does not make a mask cheap — it makes a mask less dear than the same mask pointed the other way. Confinement is what costs; aim reduces that cost and never reverses it. So there is no mask in this family that credit assignment can afford, and the line closes on arithmetic rather than on anything about the creature.

3.8 Teaching does not survive the gesture: the oscillator that gives the creature two postures costs four fifths of the lesson

A free-running two-phase vocal gesture was built here and replicated on two wiring families: produced F1 follows the alternation at 30–44 SE, a 136 ms phase at 3.67 Hz. Whether a reward could be aimed at one phase of it was then asked and refused. The refusal is reported in §2.16's terms; **what the positive control added to it is a second, larger result that had not been looked for.**

The question the control answers is whether a lesson works at all on a given genome: a target that does *not* change within the trial, and the drop in error from the first third of the session to the last. Run on three genomes, 18 creatures each, with the reward delay and credit window held fixed across them:

genome	half-centre	error drop, first third − last third
shipped default	off	+0.2197 ± 0.0120 (18.4 SE)
half-centre operating point	on	+0.0465 ± 0.0073 (6.3 SE)
the same, credit window 20× narrower	on	+0.0721 ± 0.0091 (8.0 SE)

Turning the gesture on costs 0.1732 ± 0.0140 of the lesson, 12.3 SE — about four fifths of it. The lesson still lands, at 6.3 SE, so this is an attenuation and not a block. The direction had been reported from the other side and called a success: the gesture *survives teaching*, with the two phases splitting rewarded time 0.50/0.50 and the oscillator never falling silent, which was the pre-flight that could have ended the line. The mirror image was never measured. **The operating point was chosen for the gesture's F1 swing and never once checked against the lesson**, and the two mechanisms this project most wants to compose turn out to be in tension at the only operating point where both are known to run.

The attribution is to the operating point as a whole rather than to mutual inhibition specifically: four genome fields differ between the first two rows, and two of them are spike-frequency adaptation, which could cost a lesson on its own. Separating them needs an adaptation-only arm and an inhibition-only arm, which is two more genomes and one run. Until that is done the honest statement names the configuration, not the mechanism.

The sharper form of the refusal is in the same table and does not need the mapping contrast at all. The phase-targeted lesson produces an error drop indistinguishable from zero on every genome, at every credit window and at both reward delays — eight cells, none above 2 SE, and not one of them positive — while a fixed-target lesson in the same sessions on the same creatures reaches 6.3 to 18.4 SE. Same reward machinery, same session length, same populations; the only difference is what the target is. Reward can move this creature's vowel, and it cannot move it *differently at different moments of one utterance*. That is a statement about where the reward lands rather than about when it arrives, which is what the delay oracle was run to distinguish and did.

One number in the table is suggestive and below the bar, recorded as a lead rather than a finding: narrowing the credit window twentyfold *improves* the fixed-target lesson under the oscillator, +0.0256 ± 0.0117, 2.2 SE, despite delivering strictly less total credit (61% of the trace surviving the delay against 98%). A trace spanning several phases credits all of them indiscriminately, so a shorter one should write more coherently even while writing less — which is a prediction, not a result, at 2.2 SE.

3.9 The generator and the lesson trade against each other monotonically, and the gesture is unchanged by teaching — the composition problem is all on the lesson's side

§3.8 measured that switching on a self-generated two-phase vocal gesture costs four fifths of a fixed-target lesson, at an operating point that had been tuned for the gesture's formant swing with the lesson never checked. The composition question follows: is there any point in the oscillating regime where both work? The strength of the mutual inhibition was swept and **both** quantities read at every point — the lesson from a fixed-target arm, the gesture from a phase arm — 18 creatures per arm, reward delay and credit window held fixed.

inhibition gain	lesson, error first third − last third	gesture, phase separation	phase share
0.15	+0.2062 ± 0.0079	+3.2 ± 0.1 Hz	0.49
0.30	+0.1835 ± 0.0062	+3.6 ± 0.1 Hz	0.50
0.45	+0.1243 ± 0.0095	+8.0 ± 0.1 Hz	0.49
0.60	+0.0465 ± 0.0073	+19.5 ± 0.3 Hz	0.50

Monotone on both axes, in opposite directions, every point tight. The lesson's steps are −0.0227 ± 0.0100, −0.0592 ± 0.0113 and −0.0778 ± 0.0120 — 2.3, 5.2 and 6.5 SE, accelerating. The alternation is live and balanced throughout and the voice is never silenced, so no row is a bound on the oscillating window rather than a measurement. The highest-gain row reproduces an earlier independent measurement of the same quantity to the last printed digit, which is what licenses comparing across the two runs at all.

The proposed mechanism is confirmed as a gradient, and it corrects §3.8 in the creature's favour. The account was that mutual inhibition makes the alternation an attractor that reward-driven weight changes cannot perturb; inhibition strength is therefore the knob, and it is. At the weakest gain the lesson is **statistically identical to having no gesture at all** — 0.0135 ± 0.0144, 0.9 SE from the reference. So the four-fifths cost is not intrinsic to having a generator. A creature with a live, balanced two-phase gesture can be taught exactly as well as one without; what costs the lesson is how hard the competition is driven.

One thing this run raised and a follow-up refuted, recorded with the retraction intact. The taught

separations above are 0.9–5.2% of the 375 Hz span the alternation covers when it runs free, and an earlier free-running measurement of the same gesture reported 135.6 Hz of swing. That invited the reading that the gesture *survives teaching in existence but not in amplitude* — that the phase share and the never-silent oscillator, which is all "the gesture survives teaching" ever rested on, were blind to the gesture being flattened. **That reading was wrong, and the way it was wrong is the point: the two numbers were never the same quantity.** A free-running excursion and a mean formant difference over a reward window are different estimators, and comparing them is the error, not a finding about the creature.

The matched control settles it. A third arm per gain: same genome, same seeds, same phase target mode, same accumulation, and the caregiver *muted* — praise and scold both zero, the feedback path still running on the same clock and carrying nothing. That is exact rather than approximate, because the per-phase accumulator fills on a time window while the creature is voicing and not on reward events, so the muted arm is measured by bit-for-bit the estimator the taught arms use. Free minus taught, paired on the same creature:

inhibition gain	taught	free-running	free − taught
0.15	3.2 Hz	2.9 Hz	−0.3 ± 0.1 Hz (4.1 SE)
0.30	3.6 Hz	3.5 Hz	−0.1 ± 0.1 Hz (1.6 SE)
0.45	8.0 Hz	7.9 Hz	−0.0 ± 0.1 Hz (0.2 SE)
0.60	19.5 Hz	20.1 Hz	+0.6 ± 0.3 Hz (2.1 SE)

Teaching does not flatten the gesture at any gain. The residuals are 0.0–9.4% of the amplitude they sit on and they *change sign* across the sweep, with the two largest pointing in opposite directions — the shape of tiny systematic differences measured at high precision, not of a mechanism. So the earlier report was right and the criticism of it was the error: the alternation survives teaching in amplitude as well as in existence, and **the composition problem is entirely on the lesson's side of the pair.** That is a cleaner statement than the one it replaces, and it was bought by one extra arm.

Two disciplines, one of which failed and is recorded for it. The run's printed verdict is knife-edge on pre-registered bars that happen to cross at the same point, so the trade is reported as a curve with the thresholds named as the author's choice. But the amplitude test was written *one-sided* — it asked only whether teaching *suppressed* the gesture — so the gain-0.15 cell, significant at 4.1 SE in the *opposite* direction, printed underneath a headline saying no effect was found. Technically true and materially misleading, and it is the same family as a criterion only the best outcome can fail. **A directional success test hides a significant result in the other direction, and should be two-sided unless the reverse outcome is genuinely uninterpretable.**

3.10 Credit assignment closes by oracle: handed a phase-specific place to write, the lesson still does not aim the gesture

Three candidate obstacles had been cleared in turn, each by its own measurement. The credit *pathway*: an oracle teacher reacting in 50 ms, with a credit window of 100 ms against a 136 ms movement phase, satisfied resolution and delivery together for the first time and the mapping contrast stayed null. The *generator*: teaching leaves its amplitude intact to within a few per cent of its free-running value on the same estimator. The *operating point*: a sweep of the competition strength found a setting where the lesson is statistically indistinguishable from having no generator at all. What remained was the readout — one rate-

weighted centroid over one population, read identically at every instant, so a scalar reward has no way to say *this end, now*.

So that was handed over too. Reward-driven credit was confined, tick by tick, to the half of the population that was **currently winning**, with the losing half blocked; the gating signal is the mechanism's own phase, read by the host. 18 creatures per arm.

credit block	**identity − swapped contrast**	**SE**
phase-aligned	+0.9 ± 0.2 Hz	4.3
matched, unaligned	−0.0 ± 0.2 Hz	0.2
what alignment buys	**+1.0 ± 0.3 Hz**	**3.3**

A taught separation is negative in this contrast, because the identity mapping asks the high-formant posture for the low target. The aligned result is *positive*. **The hypothesis is therefore refused on sign rather than left unproven on power**, and the closure does not depend on explaining the sign. The positive control passed at 8.0 SE and the vacuity control behaved: on a genome without the mechanism, every arm reports no phase at all.

Why this is the strongest closure this project can obtain. Every part the earlier refusals pointed at has now been given to the creature as an oracle — the lesson demonstrably lands, the generator is live and balanced, the credit window resolves a phase, the reward arrives inside one, and credit reaches only the population that is active. The two mappings remain the same experiment with the labels permuted. **No cheaper mechanism beats a perfect one**, so nothing in the credit-assignment family is worth building, including the delay line and the synaptic tag — which had already been refused from the opposite direction, because a fast gate carries no target. What is untouched is the readout itself, and two targets at two instants may require two readouts: a change to what the effector *is* rather than to how it is taught.

The control is the entire design, and one part of the matching is exact while another is impossible. Confining credit inside this population was already measured to cost the lesson a great deal on its own, so "the masked arm is worse" would say nothing. The control therefore blocks the same half, the same fraction of the time, switching on a derived period equal to one phase — same size, same duty, same switching rate, alignment broken — and it returns a clean null at 0.2 SE. Nominal credit volume is matched exactly: at every update both modes block one half and credit the other. *Effective* credit is not, because eligibility sits in the active half and alignment is precisely what puts credit there, so the aligned arm delivers roughly twice as much usable credit. **That is not a confound to remove. Alignment is credit landing where eligibility is; the two are one quantity and cannot be separated.** It is disclosed rather than corrected, which is the honest treatment of a matching that is impossible in principle rather than merely absent.

Two things recorded against the author. The residual — 5.2% of the separation it sits on, tight, with a clean matched control beside it — is a **reversal with no account on the table**. Crediting the current winner is positive feedback on the alternation and the separations do rise under both aligned arms, but why that should favour the identity mapping specifically is unexplained, and the standing rule is that a reversal is reported and nothing is built on it. And the verdict branch that printed it was itself defective: it reported the anomaly and never stated that the hypothesis had failed, putting the curiosity above the conclusion. **A positive result refutes a directional hypothesis as firmly as a null does, and a reversal branch that omits the refutation buries the finding under the anomaly.** Same family as the one-sided test of §3.9, one run earlier.

3.11 Orthogonality and instant-specificity are mutually exclusive, and the phase-specific degree of freedom belongs to the generator rather than to the lesson

Every refused attempt to teach two moments of one self-generated movement had asked for two values of *one* output dimension. A separate, positive result sat beside it: two lessons on **orthogonal** dimensions coexist where same-dimension ones collide, 0.84 retained against 0.22, and the two dimensions are carried by two separate population-coded groups. So the two moments were given orthogonal lessons — one dimension asked of one phase, the other of the other — against a same-dimension baseline reproduced in the same run.

The design is flawed, and the flaw turns out to be structural. An orthogonal pair admits a *phase-independent* solution: satisfying both targets at all times satisfies each phase, so nothing ever required the creature to behave differently at two instants. Its null therefore cannot answer the question it was built for. And there is no patched version. **If two lessons do not conflict, both can be satisfied at once and no instant-specific output is needed; if they do conflict, they conflict on one dimension, which is the baseline arm.** Orthogonality and instant-specificity exclude each other by construction, so this route closes by argument rather than by measurement — a thing that cost one run and should have cost none.

What the run found instead is better than what it was built for. The orthogonal arms show a lesson effect of +0.0614 and +0.0670 (both past 8 SE) where the same-dimension arms show −0.0017 and +0.0001 — zero — with both pairs scored on single-dimension errors so the comparison is like for like, and the orthogonal arms effectively matching the ordinary fixed-target control at +0.0721. **A lesson with a phase-independent solution is learned as well as any lesson. One without it is not learned at all: not partially, not slowly.**

The reason lies in the preceding run. The only way to reduce error on a *conflicting* pair is to change the **difference** between the two phases' outputs, and that difference was measured to be set by the competition strength — moving sixfold across its sweep — and **invariant to teaching**, with taught and free-running values agreeing to within a few per cent at every setting and the residuals changing sign. **The phase-specific degree of freedom is owned by the oscillator and is not plastic.** So the creature learns everything a phase lesson asks that can be satisfied without phase-specific output and nothing that requires it — not because reward cannot reach the right place or arrive at the right moment, both of which were handed over as oracles, but because the quantity the lesson would have to move is produced by a mechanism that reward does not write to. That is a mechanism for the whole refusal line, and it rests on two independent runs rather than on an argument.

A retraction nearly published, and the one question that stopped it. With a log-ratio error and two targets straddling the creature's output, the *summed* error is exactly constant — verified to four decimals — and the two per-phase gradients are equal and opposite. That appeared to show the design carried 3% of an ordinary lesson's drive, that the preceding runs had been judged against bars thirty times too tight, and that §3.10's closure had to be withdrawn. **It was the wrong quantity.** The cancellation is on the *mean*; the experiment measures the *difference*, where the same two gradients *add* to twice a single lesson's drive. The closure stands untouched. The check was one question — *which quantity does this experiment actually measure?* — asked before writing rather than after. **A false retraction costs more than a false finding: it discards a valid result and sends the next several runs back over ground already covered.**

4. Leverage and the geometry of the motor pool

4.1 The lesson silences the wrong side

Measured rather than fitted: a low formant target is reached by *suppressing* the upper half of the group, and a high target by suppressing the lower half — mirror-image profiles, both suppression. An earlier account in which the lesson raised the half it needed is a decisive reversal, and the reversal is what the rest of this section is an account of.

4.2 The readout is a centroid, so output potency is *signed* distance from it

The formant the host reads off a motor group is not a mean of rates. It is a rate-weighted centroid on a labelled line: neuron *i* carries a fixed position $p_i = (i - \text{begin} + ½) / n$ and the output is $\Sigma(r_i p_i) / \Sigma r_i$. Differentiating gives the influence of one neuron on behaviour with no free parameter:

$$\partial F / \partial r_i = (p_i - c) / \Sigma r_j \qquad c = \text{the output currently produced}$$

So potency is distance from the centroid the group is *currently* at — zero for a neuron sitting on it, largest at the ends, and state-dependent, moving as c moves. That immediately explains a documented out-of-sample failure: a leverage law fitted to three single-neuron costs with one free parameter reproduced them to 1.68 percentage points and then failed two half-block costs. The fit was a function of *position alone*, which is adequate while c barely moves — as it does not between single-neuron blocks — and wrong once a whole half of the group is blocked and c moves a long way. Influence is not a property of a neuron; it is a property of a neuron and the current state.

The magnitude $|p_i - c|$ is nevertheless refuted, by a run that was already on disk when the derivative was written down. One neuron blocked at three positions, 36 creatures: the top ($p = 0.964$) costs 30% of the lesson at −5.8 SE, the middle (0.464) costs 7% at −1.1 SE, the bottom (0.036) costs 9% at −1.4 SE. Middle and bottom are both null and indistinguishable from each other, and no single centroid makes a symmetric magnitude produce that: at $c = 0.093$ it requires the middle to cost about six times the bottom, and at c ≈ 0.5 it requires the bottom to cost like the top.

Potency is $(p_i - c)$ taken with the sign of the required direction of travel. A lesson that must lower the centroid can only do so by suppressing high-position neurons, so the far end *on that side* carries the lever; the centroid itself has no lever in either direction; and the opposite end is on the wrong side, because suppressing it would move the output the wrong way, so the lesson was never going to use it.

The qualitative half of this was written before the run that measured it, in the pre-registration of that experiment: index is position, a weighted mean is least sensitive near its own centre, and the end the lesson travels toward should matter more. What is new is the derivative itself, the state-dependence, and the sign — and the first arithmetic written from it, the symmetric one, was wrong.

4.3 Exactly one neuron of fourteen is expensive, and which one depends on the target

The prediction the signed account makes and a symmetric one cannot: changing where the lesson is going must *swap* which end is dear, not grade it. Three lesson targets, 36 creatures each, each arm paired on seed against its own unblocked control:

Target	p	Travel	Lever	Cost of the lever	Middle	Far end
320 Hz	0.093	0.407	**top**	**30.0% ± 5.1%**	7%	9%
475 Hz	0.300	0.200	**top**	**25.0% ± 5.9%**	6%	5%
850 Hz	0.800	0.300	**bottom**	**22.1% ± 4.4%**	4%	3%

Cost as a percentage of the lesson learned by the unblocked control at the same target. Both end columns are the single neuron at that end of the group of fourteen.

Three targets; three times the lever is the single neuron at the end the lesson must *suppress*; three times the middle is free. Every dear arm clears 4 SE and every cheap arm sits under 1.5 SE at comparable standard errors, so the cheapness is measured rather than merely unresolved. The two low targets and the high one place the lever at opposite ends of the same group, which is the swap rather than a gradient.

Both later runs carried the original target's four arms unchanged, and its headline cost reproduced to four decimals (−0.0503 ± 0.0086, −5.8 SE) in each. The runs are therefore self-validating rather than resting on a cross-run comparison.

4.4 The size of the lever is not a function of how far the lesson travels

A second quantity looked as though it fell out of the same account and does not. Across the first two targets, the distance the centroid had to travel and the cost of the lever stood in ratios of 1.356 and 1.357 — agreement to 0.1% with nothing fitted. The third target refused it outright, and the costs are non-monotone in travel, so a cost proportional to distance cannot produce the ordering at all. §2.10 records the inference lesson. What survives is the direction result of §4.3 and an explicit refusal to fit the magnitude: the end the lesson must suppress carries the lever, the middle never does, and what sets the size of that lever is open.

4.5 Position decides whether extent costs anything at all

The same two axes, measured independently in the other family and on one reference lesson of +0.1676 ± 0.0073, 36 creatures. Here neurons are *frozen* rather than being the only ones rewarded, so the axis runs the other way: the group is fourteen, the lesson is a low target, and the ladder is built from each end.

Frozen	At the lever end		At the opposite end	
1 of 14	−0.0503 ± 0.0086	30.0%	—	—
2 of 14	−0.0599 ± 0.0088	35.7%	−0.0144 ± 0.0099	8.6%
4 of 14	−0.0957 ± 0.0091	57.1%	−0.0171 ± 0.0095	10.2%
7 of 14	−0.0889 ± 0.0080	53.1%	−0.0153 ± 0.0102	9.1%

Cost of freezing, paired on seed against the unfrozen control, as a fraction of that control's own lesson. The two ends at seven neurons each are complementary halves of the group.

Read down each column. **At the lever end extent costs: four neurons against one is −0.0455 ± 0.0075, 6.1 SE. At the opposite end it costs nothing at any size** — four against two is −0.0028 ± 0.0092 (0.3 SE), seven against four is +0.0019 ± 0.0087 (0.2 SE), seven against two is −0.0009 ± 0.0086 (0.1 SE). Three steps

under half a standard error, at standard errors small enough to have resolved a step the size of the one the other ladder shows, so the flatness is measured rather than unresolved. At matched extent the two ends differ by 8.6 to 8.7 SE.

So position is not one cost among two: **it decides whether the other axis exists.** Freezing more of the end the lesson must suppress costs more every time until it saturates; freezing more of the end the lesson was never going to use costs nothing, however much of it goes. §3.7 reaches the same two-axis statement from the reward-mask family by a different manipulation, and the two were measured three weeks apart.

The agreement was then tested rather than admired. This ladder's flatness at the unused end was used to predict the one cell the reward-mask family had not measured — a half and a quarter confined to the unused end should cost the same — and the prediction was registered before the run. Measured: +0.0024 ± 0.0066, 0.4 SE, at power to have resolved a tenth of the step the lever end shows. **One family of this project predicted another's unmeasured cell, through the opposite manipulation, and was right.** Nothing else here has been confirmed that way, which is the reason the geometry is treated as settled rather than as a third experiment waiting to be run.

The lever end also *saturates*, and that is the second column's own result: seven neurons against four is +0.0068 ± 0.0068, 1.0 SE. Once the top four of fourteen are frozen there is nothing further to lose, which is why the costs at four and at seven are the same number.

This corrects a claim of this project's own, which had been recorded as "the block cost is flat in the fraction blocked, so there is no mask size that wins". That rests on exactly the two points above the saturation knee — four and seven at the lever end — while the one-to-four segment of the same ladder rises from 30% to 57% at 6.1 SE. The flatness is real and local: it is the saturation of one ladder above four of fourteen, together with the whole of the other ladder sitting at zero. It is not a property of the fraction blocked, and it was drawn from a subset of a curve the project had already measured in full.

4.6 What a restriction costs has two axes: the potency of what it restricts, and how much

Stated across every plasticity restriction the project has built, with the columns measured in separate experiments:

Restriction	Selects by	Effect on learning	Effect on forgetting
Astrocytic activity gate	activity	−81%	−19%
Per-neuron reward mask, aimed correctly	position	−46%	−30%
Per-neuron reward mask, central	position	−97% (29×)	undefined
Consolidation brake on high-traffic edges	traffic	null	null

The brake is the informative row. It binds hard — mean per-edge learning rate 0.6708 against 1.0000 unbraked, reproducing an independent earlier measurement of 0.654 — and it changes neither retention (−0.6 SE, flat across every gain floor) nor learning. A mechanism that restricts a third of all plasticity and costs nothing is not a weak version of a mask that restricts half and costs 46%: it selects a *different set*. Traffic is uncorrelated with position in the group, so a traffic-selected restriction spreads across the centroid and leaves the ends free, while a position-selected one of the same size can remove the lever entirely. **Extent alone buys nothing: a restriction whose potency is zero is free however wide it is.**

Written here first as “size is the wrong axis, potency is the right one”, and that is half wrong. §3.7 measured extent at matched aim and it costs −0.0719 ± 0.0132, 5.4 SE, and §4.5 has the same two axes in the freezing family at 6.1 SE against 0.1 SE. The statement that survives all three is: **potency decides whether extent costs anything, and given nonzero potency extent then costs until it saturates.** The brake is the potency-zero end of it and the mask ladder is the other; neither is the whole.

The first three rows are a quantitative form of the stability–plasticity dilemma in one system — every restriction cut learning more than it cut forgetting — and the fourth is not an exception to the ordering so much as a case where the dilemma never engaged. This project predicted the opposite for the brake before running it, which §2.11 records.

4.7 The larynx reads nine scalars

The motor interface is nine scalar groups, and the object identity being taught sits at the null in all eighteen vocal measurements taken. Reordering the vocal population by its own response difference lifts discriminability, so the information is present in the slice and absent from the readout.

5. The voice and the body

5.1 The voice drones

There is no syllabic rhythm in the produced voice. A +9.99 dB modulation peak that indicated otherwise was a fit on the knee of a low-pass filter, not a spectral feature of the signal. Relatedly, a 3 Hz ring in the voice is the product of two filters — a fast rate constant and a gate smoothing term — and may be no oscillator at all; it scales threefold with self-hearing, and a deaf creature still rings at +3.1 SE, so self-audition is a gain on it rather than its cause.

5.2 The frame is a body part

A single finding that explains five separate refusals at once: the apparent syllabic frame is a resonant jaw at about 4.5 Hz, a property of the articulator rather than a central clock. Three independent routes from a frame to content are closed by it, and a jaw-gate mechanism that appeared to pass both halves of its test turned out to have a mis-set anchor, with the creature’s own silences outperforming the imposed gate.

5.3 A blind controller beats feedback

A controller with no access to the sensed formant outperforms one with it: reach is immune to loop delay and a 905 ms loop is not. The precision advantage claimed for feedback was cleared by a sensorless controller, and an uninformative feature leaves only the intercept, which is the blind controller in disguise. Altered auditory feedback moves production by +0.01.

5.4 Intrinsic plasticity is the backlash

Formant actuator hysteresis measures 107 Hz with intrinsic plasticity enabled and 2.0 Hz with it off: the plant moves under its own controller. The backlash is per-neuron — pooling the regulator takes hysteresis from 107 Hz to 7 Hz and gives back 30 Hz of reach.

5.5 The ear hears its own formant and does not use it

The self-produced first formant is present in the auditory representation (+0.124 and +0.114 on two seed families) and unused by the motor system. The ear's estimate is well described by a fixed attenuation plus noise, with the attenuation free and the noise 56 Hz input-referred; the signal is localised, one frequency bin of 32 carrying 92% of the available benefit, which is destroyed by pooling. So the failure of closed-loop control here is a missing *mapping*, not a missing signal.

6. Representation and capacity

6.1 Vocabulary does not collapse, and nine words do not cost naming

Advantage over chance *rises* with vocabulary size rather than falling. At nine words against two six-word arrangements, total naming excess is conserved (-0.012 ± 0.019 and $+0.020 \pm 0.018$, opposite-sign nulls at $n = 36$). There is one real cost and it is not naming: delivered formant separation falls by 11.8 ± 3.5 Hz at matched trials per word, 3.3 SE. The shape of the word arrangement does not matter (-0.032 ± 0.027), so there is no preferred axis per creature.

6.2 The ceiling is an exponent

Delivered formant change scales with the aligned component of the learned bias as a power law with exponent about 0.61. Moving delivery by a factor requires moving the bias by a larger factor, which closes the naming line quantitatively rather than by argument: a commitment brake moved the aligned bias 1.07× where 6.2× was needed.

6.3 The derived index gap: two different results, and the boundary was the cause of the one that mattered

The creature derives its own context from an auditory rate average, and the resulting index is reliably *worse* when the teaching target tracks the heard word than when it is fixed, on 18 of 18 creatures — a gap of -0.056 ± 0.009, 6.0 SE. Three candidate causes were tested and all three are dead: gate corruption has the wrong sign; silencing self-audition does not close the gap but *widens* it by 0.037 ± 0.015, under a test with power to see a closure; and a drift account is excluded both by its own point prediction, 3.9 SE away, and by the learning arm decaying least.

Decomposed against the occupancy ceiling of §2.13, the 18-of-18 result is two unrelated effects. For the auditory-rate source the two conditions have ceilings within 0.008 of each other, so essentially the whole −0.056 is classification error. For the running-average source the ceilings differ by 0.055 while the classification difference is 0.001 — both of its arms are pinned to their ceilings, so its identically-sized gap is *entirely* a matter of one partition being more lopsided than the other, with no confusion difference in it at all. The gap is real in both; it is the same number for different reasons, and no single account can be correct for both.

What remains unexplained is therefore narrower than the original statement: one classification gap, in one source, between two arms that share the clustering rule, the feature that rule reads, and the distribution of that feature. **Any candidate that is a property of the rule is excluded structurally** — it would shift both

arms equally and cannot produce a difference between them — which retires two of the candidates on the list without measuring either.

One candidate survives that filter, and measured by intervention it accounts for the whole of the remaining gap. The window in which the partition accumulates its feature, and the moment it commits an update, are decided by whether the creature's *own larynx* is below its setpoint — a deliberate choice, since vocal silence marks listening where auditory activity does not. Because that boundary is an internal read of motor rate, silencing self-audition never touched it: the intervention that refused the self-contamination account cut the acoustic path only.

Handing the boundary to the host — the caregiver's own flag for when it is speaking, with the feature, the clustering rule and the latch all unchanged — **takes classification error to exactly zero.** All four arms built that way sit precisely on their own occupancy ceilings, under both an alternating and a shuffled word order, so the margin-constrained score of §2.13 is attained rather than merely approached. The classification gap closes completely, −0.061 ± 0.011 at 5.5 SE, under a test whose power to see a closure was checked first. **So the feature is exonerated and so is the clustering rule; the boundary was the whole of it.**

The mechanism had been sitting in the kernel's own commentary for a month, unconnected to the gap: a fragment holding only silence writes a prototype from silence and overwrites the latch. The larynx gate fires several ragged episodes per trial and some are such fragments. That also explains why the effect was a *difference* rather than a level: in the condition that produces a word-dependent voice, the fragmentation is word-correlated, while in the control it is word-independent noise.

What remains, once confusion is gone, is a **decay rather than a standing imbalance**, and the clean boundary is what made it visible. Under the oracle window the partition's lopsidedness is 0.606 to 0.609 across thirty-six creatures — zero variance, so not an outcome individual creatures succeed or fail at — and the index runs 0.919 in the first third of a session to 0.858 in the last, −0.060 ± 0.001 at **46.5 SE**, on every arm built that way. The arms using the creature's own boundary do not decay at all. Since classification error is zero throughout, the partition is becoming progressively more lopsided as the session proceeds, which is what a prototype whose learning rate has reached zero looks like under an input that is still moving.

That account had been tested and excluded, against its own point prediction, at 3.9 SE. It was tested on the arms using the creature's own boundary — which sit far below their ceiling with a fifth of the score lost to classification error, and so have no headroom in which a decay could appear. **The account was not wrong; it was measured in the one condition that could not exhibit it.** This is the reference-point error of §2.1 and §2.3 one level up: not the wrong baseline, the wrong regime. The transferable form is that a null is only as strong as the headroom the measurement had, and headroom is a quantity that can be computed in advance — §2.13's ceiling is exactly such a computation.

And the boundary is worth the whole of the behaviour, not just the carrier. The quantity the context machinery exists for is how far the creature moves its first formant *per word*. With its own derived index and a clean boundary it delivers 76.8 ± 5.0 Hz against a control arm's 23.4, a conditional effect of +53.4 ± 5.6 Hz at 9.6 SE where the creature's own boundary gives +12.0 ± 4.5. Measured against a creature that is simply *told* which word it heard, the difference is −1.0 ± 6.0 Hz — statistically identical, where the same comparison without the clean boundary is −46.9 ± 5.0, 9.4 SE below. **So knowing which word it is buys nothing beyond knowing when a word is happening.** The matched-marginal control sits below the no-

index arm, so this is word-specific delivery rather than a general increase in vocal motion.

That is a trade of oracles rather than the removal of one, and it is worth being exact about: an expensive oracle, the word's identity, has been replaced by a cheap one, its timing, at no measured cost. The cheap one is also the one the system has a real detector for — the creature goes nearly silent while it listens, voiced fraction 0.276 to 0.010, which is shipped and measured. What fails is not the signal but the gate on it: an instantaneous threshold on a noisy mean, with no hysteresis, which chatters and splits one word into several episodes of which some contain only silence.

Two things this does not say. The boundary was handed over by an *oracle*, so what is measured is the size of the prize for building a boundary detector rather than a detector — and §6.5 reports what happened when four detectors were built to collect it. And the decay above is named but not fixed: nothing here shows that restoring the prototype's learning rate recovers the 0.060 — indeed the behaviour reaches the oracle's level while still decaying, which bounds how much that 0.060 can be costing.

Two side results survived that line. Deafening improves the index in the control arm only (+0.030 ± 0.004, 7.1 SE), and makes the tracking arm's partition more lopsided (+0.066 ± 0.018, 3.7 SE). Separately, the derived context carrier is refuted directly: the index never separates the two lessons (0.23 against 0.23), while the auditory feature it is built from carries the word at 97.8% on held-out trials where the reward lands. The feature is fine; the extraction is not.

6.4 Two percent of a percent of the ticks does the work

0.2% of ticks account for the learning, and no schedule reproduces it at 3.3 SE over 32 creatures. The result carries no lesson about *which* ticks, which is stated here because it is the kind of finding that invites one.

6.5 The prize in 6.3 was for a detector, and no detector collects it

§6.3 priced the word boundary by handing it over as an oracle, and said in terms that what it measured was *the size of the prize for building a boundary detector rather than a detector*. Four detector shapes have now been built and measured side by side at full budget — 14 arms, 36 creatures, 3.4M ticks — and the prize is not collectable by any of them.

arm	dF1 (Hz)	paired vs its control	index	occupancy	slack to ceiling
handed the word (oracle)	80.7 ± 5.0	—	1.000	0.500	0.000
handed the boundary	74.6 ± 5.5	+49.2 ± 6.0 (8.3 SE)	0.892	0.608	0.000
no detector at all (EMA)	**63.5 ± 4.4**	+38.6 ± 5.1 (7.6 SE)	0.854	0.641	0.006
envelope-derivative gate	57.0 ± 6.2	+24.1 ± 7.9 (3.1 SE)	0.810	0.636	0.054
split gate	39.9 ± 3.7	+15.5 ± 4.3 (3.6 SE)	0.777	0.604	0.119
conjunction gate	39.6 ± 3.7	+16.4 ± 4.7 (3.5 SE)	0.760	0.612	0.128
larynx gate	27.5 ± 3.4	+7.1 ± 4.7 (1.5 SE)	0.694	0.642	0.164
no index	25.1 ± 3.5	—	0.000	0.000	—

The envelope gate works and is not worth having. It is the detector the project had argued itself into: the larynx gate chatters, so a gate keyed on the rising edge of the heard envelope should be cleaner, and it is — it beats the larynx gate by +29.5 ± 7.1 Hz, 4.2 SE. But against the arm that uses no boundary detector at all, reading the context index straight off a decaying average of the ear's own rate, it is −6.4 ± 7.7 — indistinguishable, on the wrong side, and ahead on only 15 of 36 creatures. The conjunction gate (heard onset *and* a quiet larynx) and the split gate (ear for the boundary, larynx for per-tick trust) both beat the larynx gate by 3.1 SE paired (+12.1 ± 3.9 and +12.4 ± 4.0 Hz, 26 of 36 each), so both clear this project's bar against the shipped gate and neither is mechanically broken — and both land roughly 24 Hz below the detector-free arm. **Nothing in the family beats the member with no detector in it.** Every arm runs on the same creatures, the per-job seed being independent of the arm, so each gap is available paired as well; that is where the sign counts come from.

Two results from the same run are not closures, and one of them replicates §6.3 on a second wiring family. That the word's *identity* buys nothing beyond its *timing* holds again here: handing over the word rather than the boundary is worth +6.1 ± 6.1 Hz, 1.0 SE, on 18 of 36 creatures — a sign count exactly at chance, measured on creatures the original claim never touched. And real headroom remains: the full oracle beats the detector-free arm by +17.2 ± 5.1 Hz (3.3 SE, 25 of 36), above the bar. **What is refused is building a detector to close that gap, not the gap itself.** The three figures are mutually consistent — 17.2 is 6.1 plus 11.1 — with the split between them poorly resolved.

The last column says why, and it is the more durable half of the result. For two balanced words the index match cannot exceed `1.5 − occupancy`, where occupancy is the share of trials falling in the busier half of the table; *slack* is the accuracy an arm still has available. The detector-free arm's slack is **0.006** and the handed-boundary arm's is **0.000**. **The leading arms are already as accurate as their own partitions permit, and every arm in the table lands where its ceiling puts it.** The arms that lose are exactly the arms sitting below their ceilings. So the open problem is not *when does a word happen* but *where to put the threshold so the trials divide evenly* — a different question, and one no gate in this family was built to answer.

Three things this does not say. Splitting the trials evenly is shown to be *necessary* for a derived index to beat the detector-free arm, not sufficient: for an arm at its ceiling `index ≡ 1.5 − occupancy` identically, so accuracy and balance are the same number and this run cannot separate them. The boundary prize itself is genome-dependent — handed-boundary minus detector-free is +11.1 ± 7.1 Hz here and +27.4 ± 7.0 on the earlier family, and the two runs share no creatures, so §6.3's figure and this one are two samples of a range rather than a disagreement. And the two gate shapes built for this run were very nearly refused on a short smoke that mis-measured them: their gate-events-per-trial read 0.55 and 0.59 against a design of 1.0 on a 17×-short run and 1.76 and 1.86 at full budget, because a rate-thresholded gate's behaviour depends on a rate distribution that is still settling in the first few percent of a creature's life. Only a handed window was budget-invariant.

6.6 Saturation identifies what a matched comparison cannot

§6.5 left a question it could not answer. The leading arms sat at the ceiling `index ≤ 1.5 − occupancy`, and at that ceiling the two quantities are *identically equal*, so no comparison among those arms can say whether the index is limited by its accuracy or by how evenly its partition divides the episodes. What was needed was a lever that moves occupancy while leaving the feature untouched.

DeSieno's conscience is one: it adds `(slots × win_fraction − 1) × dscale` to each slot's distance,

penalising a slot that wins more than its share, with `dscale` the running mean winner distance so the penalty is in the units of the distance it joins. That term had an implicit gain of 1. Scaling it leaves the feature, the prototypes and the learning rule alone and changes only the pressure toward an even split.

gain	index	occupancy	ceiling	slack	dF1
shipped (no conscience here)	0.854	0.641	0.859	0.005	63.5
1	0.887	0.605	0.895	0.008	66.6
2	0.890	0.522	0.978	**0.088**	71.8
4	0.874	0.515	0.985	**0.111**	67.0
8	**0.520**	0.502	0.998	**0.478**	22.1

Pushing the ceiling past the plausible range separates the two by making the index fail to follow. Occupancy goes from 0.641 to 0.502 — a perfectly even split — and the ceiling with it to 0.998, while the index saturates at about 0.89 and the slack opens from 0.005 to 0.478. **The partition's balance was binding only while occupancy exceeded roughly 0.52; past that the limit is the feature.** No measurement taken while the index sat on its ceiling could have established this, and the method is general: when a quantity is pinned to a bound, drive the bound beyond the range of interest and see whether the quantity tracks it.

A perfectly balanced partition is not an informative one. At the highest gain the table is balanced to 0.502 and the index reads 0.520 against a chance level of 0.500 — the conscience has overwhelmed the data and is assigning slots by turn-taking rather than by the word. Delivered dF1 falls to 22.1, *below* the arm with no index at all, and paired against the shipped creature that is −30.4 ± 5.6 (5.4 SE). This was pre-registered as one of three admissible outcomes before the sweep ran.

The behavioural question was then re-asked on creatures the claim had not seen, by offsetting the genome seed so that a fresh run's creatures are the next block of the same family. The mechanism replicated to three decimals: two halves starting from different places (occupancy 0.641 and 0.653) land on the same one (0.522 and 0.523), and the treated arm delivers 71.8 and 71.9 Hz in the two families against an oracle at 80.6 and 80.8. **The noisy quantity is the untreated baseline, not the treatment.**

The behavioural gain did *not* at first clear this project's bar and was recorded as such: the registered contrast pooled to +12.7 ± 4.7 Hz (2.7 SE) over 72 creatures, 45 of them positive, where the threshold is 3. Roughly eighteen more creatures would have closed the gap arithmetically and none were run, because adding samples until a number crosses a threshold is optional stopping.

6.7 The same effect, settled at a sample size chosen in advance

An effect at 2.7 SE cannot be claimed and need not be abandoned. The alternative to adding creatures until it crosses is to fix the sample size from a power calculation and write the stopping rule down first. Before the run, the following were committed in source: two further blocks of 36 creatures (the genome seed walks the same family forward in blocks of 36), **n = 144 exactly**, the 3.0 SE bar, and a rule stating that a failure *ends the line* with 2.5–3.0 SE counting as a failure. The power limit was stated too: with a per-creature SD near 40 Hz the smallest effect certifiable at 144 is about +10 Hz, so a refusal would have meant “under about ten hertz” and never “zero”.

block	creatures	contrast
1, discovery	0–35	+14.81 ± 5.85
2, fresh	36–71	+10.51 ± 7.36
3, new	72–107	**+15.02 ± 6.71**
4, new	108–143	**+20.53 ± 8.50**
pooled	**144**	**+15.22 ± 3.56 — 4.3 SE, 94/144**

It clears. All four blocks are positive, so the pre-registered sign-change refusal never fired, and the two blocks the claim had never seen are the *strongest* of the four — the opposite of the shape a discovery-sample artefact makes. The sign test agrees without any distributional assumption: 94 of 144, **z = +3.67**. The per-creature SD came in at 42.7 Hz against the 39.9 predicted beforehand, so the calculation that chose the sample size was sound rather than lucky.

And 85% of it is in the voice. The statistic is a difference of differences, so a sagging control inflates it — which is what the first half looked like. Over the full sample it decomposes into the treated arm rising **+13.00 ± 3.10 (4.2 SE, 91/144)** and the matched-marginal control falling only −2.21 ± 2.16 (1.0 SE, indistinguishable from zero). The voice arm clears the bar on its own.

What it collects is the gap §6.5 left unclaimed. The distance from the derived index to the host-written oracle index goes from +19.16 ± 3.18 (6.0 SE) to **+6.15 ± 2.78 (2.2 SE)**: the treatment closes +13.00 ± 3.10, about two thirds of it, and what remains is no longer clearly resolvable. Four boundary detectors were refused against that gap and balancing the table was refused against it; this collects most of it by sharpening the partition instead. The specificity control is exact — the oracle arm is **bit-identical across the two genomes on all 144 creatures**, a jackknife standard error of precisely zero on an arm the mechanism cannot reach.

The rule that made the result credible is the one that was never invoked: that a 2.9 SE outcome would have ended the line. A pre-registration only constrains if it would have refused something, and this one would have. One audit is owed and is now safe to run precisely because the claim no longer rests on it — block 1's baseline came from a twenty-arm log rather than a dedicated run, and blocks 2–4 alone pool to +15.35 ± 4.34 (3.5 SE) and clear without it. Swapping that baseline was deliberately not folded into the settle, because changing a comparison after seeing the number is the flexibility the exercise exists to remove.

6.8 Consistency at low power is not a collapse, and it refuted the opposite claim too

The creature's two-phase gesture and its taught vowel do not compose, and that was established over four generations and twelve cells of a two-parameter plane — the inhibition strength, the tonic drive, the interval where the first two sweeps crossed, and finally the untested corner where weak inhibition meets low drive. The last of those is in §9's table. What is recorded here is a reasoning error made while writing it up, caught before it reached the record.

Two pairs of cells reach nearly the same gesture amplitude by *different* knobs, and in both pairs the lesson agrees:

same gesture, different route	lesson	agreement
amplitude 9.4 Hz via drive vs 10.0 Hz via gain	0.0871 vs 0.0796	0.7 SE
amplitude 18.5 Hz via drive vs 19.5 Hz via gain	0.0627 vs 0.0465	1.7 SE

The tempting reading is that the two-parameter plane is really one-dimensional for this trade — that the lesson's cost is set by the gesture asked for and not by the knob that asks. It would have explained why three single-axis sweeps kept tracing one curve, and it is wrong, for two reasons that are worth stating generally. **First, three standard errors here span 39% and 53% of the quantity**, so a route difference of tens of per cent would be invisible; agreement at this power is not identity. **Second, the residuals inside each pair are steeper per hertz of amplitude (0.0125 and 0.0162) than the slope between any neighbouring cells on a single axis (0.0009 to 0.0051)**, so they are not the small amplitude mismatch inside the pair — they are genuine route residuals that merely cannot be resolved. The second check is the sharper one: a residual that is *steeper* than the curve it is supposed to lie on is evidence against the collapse, not for it.

The same two checks cut the other way, which is why this is a finding and not only a caution. An earlier generation had recorded the opposite claim — that the two axes are *not* one curve — on values interpolated to a common amplitude, together with fitted exponents 45% apart. A later run had already shown interpolation across an unsampled interval to be unreliable in this very sweep, predicting a pass by 0.8% where the measurement found a 3.7 SE failure. Measured at matched amplitude, the difference does not appear. **Both “the axes differ” and “the axes are the same” are unsupported; the trade is one curve to within tens of per cent, and that is all twelve cells can say.**

The rule extracted: before writing that one quantity *is a function of* another, quote three standard errors as a fraction of the quantity, and compare the residual to the slope of the curve it is claimed to lie on. And when a new analysis contradicts an older record, test its power before believing it — the tidier claim is not automatically the right one.

6.9 Supplying a perfect version of the missing mechanism, and pricing what it should have bought

An experiment here refused a phase-targeted lesson on eight configurations and its verdict named a cause: reward reaches the vocal module as a quantity with *no phase-specific target to write into*. That is a claim about a missing place to write rather than about timing, and it is the reason a delay line and a synaptic tag were both refused without being built. The creature already has a mechanism that provides a target per context — a bias table indexed by a context signal, which works whenever the index is informative. So the cheap test is to supply a *perfect* index and ask whether the lesson lands at all.

The context was pinned, every tick, to the creature's own oscillator phase, giving reward a separate bias table per phase. The credit window was narrowed twentyfold so the eligibility trace (~100 ms) fits inside one phase (~136 ms); without that a phase-specific *place* to write would receive phase-*mixed* content and a null would be unreadable. Every precondition was then satisfied: context presence 1.000 on the treated arms and 0.000 on all five controls, a fixed-target lesson landing at 12.0 SE on the same genome, 60.7% of credit surviving the delay, and the gesture alive at 19–20 Hz.

The taught phase separation came back **+0.19 ± 0.35 Hz (0.5 SE)** against a 3 SE bar. Three standard errors is 1.05 Hz against a 19 Hz gesture, so the refusal means “under about one hertz”, never “zero”.

The result is not that null. It is a diagnostic that was nearly left out. A null of this shape is ambiguous in a way that matters: either reward wrote phase-specifically and the formant ignored it, or the write never happened and the mechanism never engaged. Those license opposite conclusions, and nothing in the headline number separates them. The quantity that does — the mean absolute difference between the two context tables — already existed in the shared return structure and this experiment simply did not carry it.

arm	**table divergence**	**delivered formant difference**	
word index, host-written	0.0288	80.1 Hz	the oracle
word index, derived	0.0280	74.4 Hz	the creature's own
matched-marginal control	**0.0268**	**23.5 Hz**	diverges; index uninformative
no context table	0.0000	22.5 Hz	baseline
phase index, per tick	**0.0216**	**< 1.1 Hz**	70–100 SE of divergence

The treated arms diverged by 0.0216 at 70 and 100 SE against *exactly* zero on the shared-bias arms, in opposite directions for the two mappings. **Reward did write per phase, and the formant did not follow.** The upper rows then price what that should have bought: roughly 1900–2000 Hz of formant difference per unit of divergence over the no-table baseline, so 0.0216 should be worth about **40 Hz**. A **forty-fold shortfall**, in a run that could resolve 1.05.

The third row is what makes that argument valid rather than rhetorical. It diverges as much as the working arm and delivers nothing over having no table at all, because its index does not track the word. **Divergence alone is insufficient; it has to be aligned with the distinction being drawn.** In the phase run the index is aligned by construction, so the comparison is between two aligned cases. And the obvious objection fails on arithmetic: the reward delay misdirects 50/136 = 37% of each write into the neighbouring phase's table, worth a 3.8× attenuation even if all of it subtracts.

So the obstacle is expression, not credit assignment, and that unifies with a separate closure reached the same week: twelve cells across a two-parameter plane found no operating point where the oscillator and the lesson coexist (§9). This says why. Mutual inhibition holds the formant at one of its two ends, and a bias written into the *correct* per-phase table still cannot shift it.

The wider value was avoiding a build. The songbird literature's account of temporal credit — a sparse sequential representation of time, so that each moment becomes a separate context and reward can be assigned per moment — had been the most promising unbuilt direction here, and the machinery for it (a propagating chain) already existed switched off. Its job would have been to *derive* this index. An oracle version of the same index buys under one hertz, so a derived one buys at most that. The method generalises: when a verdict names a missing mechanism, supply a perfect version of it before building a real one, instrument the perfect version so its null is interpretable, and price the outcome against a working use of the same mechanism elsewhere in the system. That is what turned “no effect” into “forty times short, and the block is downstream”.

6.10 A factorial task, stored non-factorially

The creature names four signals laid out as a 2×2 — two first-formant levels by two second-formant levels — and that layout is what made four signals work at all, since four levels along a single axis read at chance even with a perfect index. So the code it learns is, by construction, *two independent bits*. The question here is

whether the creature represents it that way.

The reason to ask was a mechanism. Every context in this architecture owns its own parameters: the stored table is indexed by the active context and nothing is shared between contexts, so a creature that learns one context perfectly has learned nothing about another even when the two share a feature. The iterated-learning literature makes that matter, because its account of why culturally transmitted codes survive turns on learners *generalising from sparse input* — and a learner with no shared parameters cannot. That looked like the explanation for why a four-signal code decays down a lineage where a two-signal one does not, and it named a fix: parameters per *feature* rather than per *context*, which would also be cheaper.

But parameters per feature can only be cheaper if what is learned is already additive, and that is measurable on a trained creature with no new mechanism at all. For a 2×2 the decomposition is exact and orthogonal, with no fitting and no free parameters: each neuron's four stored values are a mean plus three orthogonal contrasts — one per feature, and their interaction, which is the part a factored representation cannot hold. The three contrasts partition the context-dependent power exactly.

That exactness is what makes the null derived rather than chosen. Four values carry one mean and three orthogonal contrasts, so if the four were independent the power would split equally among them and the interaction's share would be one third. Chance is therefore 0.333 by construction, not by assumption.

index	interaction share	feature 1	feature 2	context-dependent
host-written	**0.325 ± 0.006**	0.335	0.340	0.78
derived	0.331 ± 0.007	0.345	0.324	0.74
derived, sharpened partition	0.333 ± 0.010	0.321	0.346	0.74

The three contrasts hold equal power. 0.335, 0.340 and 0.325 is the equipartition expected of four independent values, and the measured share sits 1.4 standard errors below chance with a standard error of 0.006. **This is a well-resolved equality with chance, not a failure to resolve a difference** — a larger sample would pin it closer to one third rather than pull it away, which is the opposite of what an underpowered null does.

The check that could have voided it passes in the useful direction: roughly three quarters of the stored power is context-dependent, so the table distinguishes its four contexts strongly. There is no shortage of content. The content simply has no feature structure.

So a factorial task is stored non-factorially, as four independent associations. The targets are a product by construction and the representation is not. With one independent parameter set per context and a learning rule that correlates reward with per-neuron perturbations, nothing pushes toward a shared representation, and none appears even when the task supplies one for free.

A sharpening mechanism that improves *which* context wins changes this not at all (0.331 to 0.333), which is consistent rather than disappointing: it is a result about partitioning the input, not about what each partition stores. Two independent measurements now say so.

The careful part is what this refuses. It refuses the *rationale*: a factored representation cannot compress structure that is absent, so "the same code for half the parameters" is not on offer, and the generational decay is not a parameter-sharing problem —

which corrects a hypothesis recorded alongside that decay the same day. It does *not* refuse the possibility that a factored learner could *impose* additivity and still solve the task, since the targets really are a product. But that is a constraint rather than a compression, far weaker than the claim the lead was built on, and the cheap evidence points against it. The method generalises: a pre-flight that costs one run can kill a build whose rationale looks solid in two separate literatures, and it works best when the quantity is chosen so that the null falls out of the algebra instead of being assumed.

6.11 A third of the creature is not needed for the task it was built for

The creature carries seven modules, and the conditional-naming task it does best is purely acoustic: a word is heard, and a word is produced. A measurement of its running cost made the mismatch concrete — the vision module holds 256 of 1166 neurons, and the context module fires at **0.00 Hz** in the shipped regime, because the derived index reads the auditory module instead and the context module is silent unless a host writes into it. Three modules and five projections looked like they could go.

What makes this a result rather than a tidy-up is that the deletion cannot be assumed harmless. Removing a module re-rolls every random draw after it, so the trimmed genome is a *different creature*, not a subset of the old one: its wiring differs everywhere, and the naming number already banked for the full genome does not transfer. Naming has to be re-established, and the full genome has to be re-run beside it under identical conditions rather than cited.

It also has to be recalibrated. Each module's target rate must equal its free-running rate or every later measurement includes homeostasis fighting the genome, and the association module had lost two of its afferent projections — it went stale by 1.15 Hz. The re-measurement is reported in §2 because it turned out not to be a measurement at all.

creature	neurons	synapses	four-word naming, pooled n=72
full	1166	37,275	+0.1316 ± 0.0180
trimmed	**782**	**21,213**	**+0.1306 ± 0.0180**

A third fewer neurons, 43% fewer synapses, and the difference in naming is −0.0010 ± 0.0255 — 0.04 standard errors. Both genomes were run on two independent families of 36 creatures each, and the full genome reproduced its own banked figure to three decimal places, which is what licenses the comparison. The trimmed creature is deterministic in its own right.

The null is stated as a bound, because a null without one says nothing. At two standard errors the trim costs at most 0.052 of naming excess against the full creature's own 0.132 — so *no loss larger than about 39% of the effect*, with a point estimate of −0.001. That is the trade on offer: it is not "free", and it should not be quoted that way.

What would have refused it: a resolved loss in the paired excess, or a creature count that fell because the quieter trimmed voice produced too few utterances to score. Neither happened — no creature was dropped on either genome.

One claim did not survive. On the first family the trimmed creature's second formant looked *better* (+0.045 ± 0.032), which was recorded as suggestive and explicitly not claimed. Re-asked on a fresh family it read +0.018 ± 0.032, and the overall advantage reversed sign. The effect was an artefact of the sample that produced the hypothesis, and labelling it as such in advance is the only reason its refusal cost nothing.

6.12 Reward supplies the separation and the lesson supplies the assignment

A double dissociation on the same 36 creatures, with each half null on the other's axis. Reward alone buys the entire produced separation and none of the mapping: a yoked arm, receiving reward in the same proportions on the same schedule but half a trial out of phase so that it cannot land on anything the creature did, gains +29.3 ± 2.8 Hz of between-word first-formant separation (2.50×, 10.6 SE) while its direction reads 0.247 against its own null of 0.249 — 0.1 SE. The lesson buys the entire mapping and no further separation: taught minus yoked is +0.1899 ± 0.0283 in direction (6.7 SE) and −4.4 ± 4.4 Hz in separation (1.0 SE). So reward makes the creature produce measurably different vowels for different words, with the pairing arbitrary, and teaching decides which word gets which vowel.

The result explains an earlier unresolvable contrast rather than merely adding to it. Taught minus yoked would not resolve on any accuracy column (1.7–1.8 SE, and 2.2 SE on separability) because those columns read distinguishability, which reward has already maximised; the lesson's contribution is invisible to them by construction and reads 6.7 SE the moment the question is asked on direction instead. The per-creature null is what keeps the yoke's 0.247 honest: it is chance because that creature's own null is 0.249, not because 0.247 looks low.

A claim retracted on the way: the opposite conclusion — that the expanded code was not reaching the formant — was first asserted from a per-creature correlation null (+0.082, 0.5 SE) while the per-arm contrast on the same column read 10.6 SE. The two measures are read off different windows, so two noisy measures of related quantities can correlate at zero while both move hugely. A null correlation between noisy measures is not a mechanistic absence.

6.13 One creature does both

Each half of the project's vocal behaviour was established on a different individual: repetition on a freshly hatched creature and four-word conditional naming on a trained one, with no individual ever asked for both. Read off the live session after the lesson, on the same creature, the echo of which word was just heard reaches 0.955 ± 0.007 against the 0.75 bar carried over from the repetition milestone, while naming reaches +0.1790 ± 0.0197 over that creature's own per-creature null (9.1 SE), at $n = 36$. The two capabilities compose in one individual.

What it cost to get right: the first pre-flight read the echo at chance on every arm that had lived a session, and the cause was the measurement passing its own negative control flag — the caregiver never speaks — into the reader. Every arm that lived a session was scored on a creature that was never spoken to, while the positive control reached the same reader by a different code path and did play words. A positive control that reaches the measurement by a different code path from the treatment arms certifies the reader and says nothing about the protocol. Two corroborating diagnostics offered alongside it, halved auditory drive and a voiced fraction down by two thirds, are both what a word that never plays produces: one fault counted three times rather than three pieces of evidence.

7. Transmission between creatures

7.1 One creature teaches another: a learned distinction crosses to a second individual through the air, and narrows on the way

Everything else in this document is one creature and a host. This is two creatures. Generation 0 is taught by the host in the shipped way; what it actually produces for each of the two words is measured; and generation 1 — *a different individual*, its own wiring and its own noise from its own seed — both **hears** those vowels and is **rewarded toward** them. No genome is passed and nothing is bred.

The design is shaped around a refusal rather than a hope. Teaching toward a *heard* target is a measured null here: reward cuts error by 24% toward a fixed target and by 0.1% toward a heard one. So "imitate what you hear" would have failed for a reason already known and said nothing about transmission. The tutor therefore supplies *both* channels, which is what being taught by someone means.

generation	**distinction produced for the two words**
0, taught by the host (which demanded 460 Hz)	+96.1 ± 5.1 Hz
1, tutored	+25.6 ± 7.6 Hz
1, the tutor's vowels with the labels **swapped**	−27.7 ± 6.5 Hz
1, the tutor's centre, contrast widened to the host's	+61.0 ± 11.3 Hz

Transmission: +53.3 ± 10.0 Hz, 5.3 SE. The two arms it is drawn from differ in nothing but which tutor vowel goes with which word — identical acoustics, identical spread — so the quantity is the *information* in the tutor's voice and not its sound. Both generations' lessons demonstrably landed, at 28.2 and 4.5 SE, so the fourth rung is satisfied before the verdict is read.

And the result is symmetric, which is stronger than the contrast alone. The swapped arm did not come back at zero. It came back at −27.7, a near mirror image of the tutored arm's +25.6. **The pupil learns an inverted mapping about as readily as the true one.** That rules out the reading where one arm is simply broken, and it is what transmission actually means: the pupil acquires whatever convention its tutor happens to hold, with no preference for the one the host originally taught.

It narrows, and most of the narrowing is the tutor being quiet rather than the channel being lossy. The host demanded a 460 Hz distinction; generation 0 produced 96.1; the tutored pupil produced 25.6 — a total shortfall of 70.5 ± 9.2 Hz across one generation. But a tutor hands on far less than it was asked for, so the pupil is being set a much narrower task than its tutor was. Restoring the host's spread around the tutor's own vowel centre recovers to 61.0, **+35.4 ± 13.6 Hz over the tutored arm (2.6 SE)**: directional, below the 3 SE bar, and not claimed as established. The residual against a host-taught creature, 35.1 ± 12.4 Hz, is 2.8 SE and likewise not claimed. **So whether anything is lost beyond the width of what the tutor says is genuinely unresolved here**, and the honest reading is that the narrowing is dominated by a quantity the experiment also measures.

Three limits, stated rather than left to be found. **This is one generation.** Whether the ratio is stable, decays to nothing, or stabilises at a narrow convention is what iterated learning is about, and the run cannot say; a second generation is the same

code with the tape taken from the pupil. **The ratios quoted above are ratios of means and are descriptive only** — the per-creature paired ratio with a jackknife is the statistic, and it was not computed, so no proportion here carries an interval. And **the tape carries formants only**: pitch stays the host's, because the creature's own f0 is not recorded. That is a limit of the recording rather than a claim that pitch does not matter.

One design error, caught by a smoke run and worth keeping. The first version handed on 621 and 620 Hz — a 0.3 Hz distinction — because the default taught arm *is* the heard-target mode this project had already measured at 0.1%. A tutor with nothing to say cannot teach, and the run would have been void by construction while looking like a clean negative. The fix was to give *both* generations the word-conditional configuration every distinction here is measured under: the tutor so it has something to hand on, and the pupil so it can structurally hold it. **Giving it only to the tutor would have tested whether a creature can learn something it has no way to represent.**

7.2 Down a chain the convention decays to nothing — but a width-limited channel with no information loss would look the same, and that is not yet separated

§7.1 established transmission between two individuals and left one generation of narrowing. One step cannot tell a **decay** that reaches zero from a **convention** settling at some narrow width, and those mean opposite things about whether anything can be held between creatures. Four steps separate them. Each generation is a different individual, taught from the previous generation's own production.

generation	faithful chain	labels flipped at every step
0 (host-taught)	+96.1 ± 5.1 Hz	shared
1	+25.6 ± 7.6 Hz	−27.7 ± 6.5 Hz
2	+10.5 ± 5.4 Hz	+13.6 ± 5.3 Hz
3	−0.3 ± 3.7 Hz	−2.3 ± 4.4 Hz
4	+0.5 ± 3.1 Hz	−2.1 ± 3.3 Hz

It decays. Generations 3 and 4 are both within 3 SE of zero. The last generation's own lesson landed at 4.3 SE, so the final zero is a creature that learned something and had nothing to learn — not one that failed to learn. **Culture starts and does not accumulate.** The first two generations reproduce the preceding experiment to the digit on the same seeds, which is what licenses reading the deeper ones at all.

What is deliberately not claimed. Two step ratios are resolved, 0.266 ± 0.084 and 0.409 ± 0.157, and they are not distinguishable from each other (0.8 SE) — so there is no claim that the loss per step is constant and none that it changes. Two points do not carry a curve. The deeper ratios are *suppressed rather than printed*, because a quotient by something indistinguishable from zero invites a reader to take a sign from noise; an earlier build of the same experiment printed one such ratio as "+4.367 ± 48.071".

A pre-registered test that came in below its bar, and why. The control was a positive prediction rather than a null: flipping the two labels at every step should make the sign *alternate* down the chain, which is far harder to produce by accident. The registered statistic — the mean of (predicted sign × magnitude) over all four generations — came in at 2.7 SE, short of 3. The per-generation signs are correct for generations 1, 2 and 3 and wrong for 4, and generation 4 is precisely where the faithful chain is already at zero: **a generation with nothing left to invert.** So the summary was diluted by its own later terms, which is a property of the

statistic chosen and not a finding. Inversion itself is not in doubt — the flipped arm at generation 1 sits 4.3 SE from zero and reproduces the previous run. **The lesson is about summary statistics: one number averaged over a series that dies halfway through is the wrong summary, and choosing it in advance does not make it the right one.**

And the decay does not establish a lossy channel. §7.1 left open whether anything is lost beyond the *width* of what a tutor says; the two contrasts that would have settled it came in at 2.6 and 2.8 SE, both below bar. A tutor hands on a narrower distinction than it was asked for, so each pupil is set a harder-to-hit task than its own tutor was — and **a width-limited channel losing no information at all produces a decaying series too.** "It decays" and "the channel leaks" are different claims and only the first is measured here. Separating them is the same code with each generation's tape widened to a fixed spread around its own tutor's centre: if the series then holds, the loss was width.

Two guards were added between the smoke run and the real one, and the first is the reason this result can be stated at all. An earlier build printed a confident "it decays" over a chain whose *generation zero had no distinction in it* — noise propagating through four generations, where "it decays" would have been the most confidently wrong sentence available. The guard now voids the run unless the first generation has something to transmit, checked before the series is interpreted rather than after. It is the same failure the preceding experiment had in its own first version, which is the argument for writing the guard into the experiment instead of remembering to look.

On the comparison with human iterated learning: artificial languages passed down chains of human learners become *more* structured, the opposite direction to this. The comparison is offered as loose and should be read that way — that quantity is structure and learnability, this one is the magnitude of a single acoustic distinction, and a magnitude that shrinks is not the same object as a grammar that regularises. What the two share is the method, not the measurement.

7.3 The decay was a width limit, not a lossy channel — and holding the width fixed turns transmission into a ratchet that locks in arbitrary conventions

§7.2 found the convention decaying to nothing down a chain and could not say whether the channel loses information or whether each tutor simply says less than it was asked, handing its pupil a smaller task. The separation: carry the tutor's own vowel centre *and its own ordering*, and re-spread the pair to the original log distance. The mapping propagates; the width is held fixed. A third chain exchanges the inherited ordering at every step, because a re-imposed width hands every pupil a wide target whatever its tutor said, and without that control a surviving series could carry nothing inherited at all.

generation	plain chain	width held, signed	width held, magnitude	ordering flipped each step
0	+96.1 ± 5.1	+96.1 ± 5.1	96.1 ± 5.1	shared
1	+25.6 ± 7.6	+61.0 ± 11.3	63.8 ± 10.4	−54.9 ± 9.7
2	+10.5 ± 5.4	+31.8 ± 20.8	88.3 ± 5.6	+55.1 ± 11.7
3	−0.3 ± 3.7	+44.2 ± 18.1	83.4 ± 5.9	−56.7 ± 14.5
4	+0.5 ± 3.1	+35.5 ± 19.7	85.6 ± 5.5	+61.9 ± 16.3

It was width. With the width held, generation 4 carries 85.6 ± 5.5 against generation 0's 96.1 ± 5.1 — a difference of 1.4 SE, **no detectable loss across four generations** — where the plain chain lost 16.0 SE

worth over the same span. The flipped chain alternates on **4 of 4** resolved generations at 55–62 Hz, so the mapping is genuinely inherited rather than manufactured by the target. **Culture here does not decay because the channel loses information; it decays because each tutor produces a narrower distinction than it was taught, and the next pupil can only aim at what it hears.** That relocates the limit from transmission to production.

And the mechanism turns out to be a ratchet, which nothing predicted. Five of the eighteen chains ended holding the *inverted* convention at full strength, and **eighteen of eighteen never changed sign after the first pupil**. The inverted chains are precisely those whose first pupil nearly failed: magnitude 5.1 ± 1.8 at generation 1 against 86.4 ± 7.5 for the rest, a difference of 10.5 SE. When a pupil lands near zero, which way round its pair came out is *noise* — and the widened tape takes that sign, restores it to full width, and hands it on, after which it never reverses. The magnitude series shows the amplification directly, rising from 63.8 at generation 1 to 88.3 at generation 2 as the near-failures are widened. **The mechanism does not merely preserve a convention; it restores the magnitude of whatever convention it finds, including one that began as noise.** An arbitrary convention arising from a near-tie and then held stably down a lineage is a recognisable phenomenon rather than a malfunction — and the sharpest available reason to distrust a single chain, since a quarter of them ended up with a different language.

The verdict this run printed was **wrong**, and the error has the same shape as the one in §7.2. It read the *signed* mean, 35.5 ± 19.7, which fails a 3 SE bar — because a chain confidently holding the inverted convention contributes a large negative number that cancels against the faithful ones. **A signed mean across units is not a measure of magnitude when the sign can differ between units**, and the mean absolute value separates "the distinction shrank" from "some units hold the opposite convention" — two different findings that the registered statistic merged into one wrong sentence. That is now twice in consecutive runs that a pre-registered statistic was the wrong *shape* for its data. **Pre-registration prevents fitting a test to the results. It does nothing about choosing a test incapable of expressing them.** No re-run was required: the data were complete and the correction is arithmetic on them, so the corrected condition was evaluated against the run's own numbers and the source fixed for next time, rather than spending two hours re-deriving numbers that were already right under a sentence that was not.

7.4 A four-signal code crosses the air, on a condition the creature derives itself

Two results here each stood and had never been put together. One: the creature names *four* signals, laid out as a 2×2 of two first-formant levels by two second-formant levels with every target inside its measured learned reach, at 5.6 and 4.0 standard errors on the two axes — using an index written by the host. Four signals along a *single* axis read at chance even with a perfect index, so the layout is what rescued it, and that makes the code two independent *bits* rather than four levels. Two: a *two*-signal convention crosses between distinct individuals, each pupil deriving the condition from its own ear, faithfully enough that a lineage loses nothing over four generations.

Three design choices carried the composition. **Generation zero was run twice** — once on a host index, once on a derived one — so that a failure would say *which* of the two unknowns failed rather than leaving the run uninterpretable. **A flipped chain was run alongside the faithful one**, with the inherited mapping exchanged at every step: the tape re-imposes the host's width on both axes, so a widened chain carries a wide code whether or not anything was inherited, and only an *alternating* sign distinguishes the two. **And that forced three generations**, because with a single step there is no alternation to observe. An earlier version of the two-signal experiment had dropped exactly this, and its flipped chain held one sign instead of alternating.

The first run failed, and named the lever.

index	bit one	bit two
host-written	+81.0 ± 4.0 (20.5 SE)	+148.8 ± 10.4 (14.4 SE)
derived	+42.1 ± 4.3 (9.9 SE)	**+12.3 ± 8.9 (1.4 SE)** — 8% of the host's

So production was fine and the limit was the derived index at four contexts. The first bit *did* cross, and was genuinely inherited — the flipped chain alternated in 17 of 18 lineages — but it decayed, losing 28.3 ± 6.6 Hz by the second generation where a two-signal code loses nothing over four. **And the second bit was not underpowered but absent.** That distinction was forced by a rule written before the run: a bit landing between two and three standard errors is underpowered and deserves a larger sample, whereas this one sat below that band. At a sample eight times larger it would have cleared three standard errors while *still* being 8% of what the host index delivers — and a bar cleared only by adding samples is not a mechanism working.

The second run switched on the one measured lever for that partition: a conscience term that penalises a slot for winning more than its fair share, at the gain settled earlier at a pre-committed sample size. Two genome fields; nothing rebuilt. **The two runs share creatures, and the host-index arm proves it** — the conscience touches only the derived path, so the host arm came back bit-identical, a maximum absolute difference of 0.0000 on both bits. That is what licenses pairing the two runs lineage by lineage.

pre-registered test	result
primary: bit two, paired difference	**+79.68 ± 15.43 (5.2 SE)** — met
secondary: bit one must not fall	−4.44 ± 5.16 (0.9 SE) — did not fall
second generation, both bits	+25.31 ± 3.81 (6.6 SE) and +75.27 ± 25.03 (3.0 SE)
inheritance: flipped chain alternates	**18 of 18 lineages on both bits**

A four-signal conditional code crosses the air, on a condition the creature derives from its own hearing, inherited rather than re-imposed, holding over two generations. The secondary bar mattered: a conscience strong enough to even out a partition is also strong enough to overwhelm the data and assign slots by turn-taking — at high gain on two contexts the index had gone to chance — so a second bit bought by wrecking the first would not have been a gain.

Three limits, stated because the bars were close. The second bit at the second generation sits *exactly* on its three-standard-error bar and must not be quoted as though it were comfortably past. The decay improved from 4.3 to 2.6 standard errors and is no longer established, but the improvement itself is 1.9 and is *also* not established, so the prediction that sharpening the partition would stop the decay is suggestive rather than shown. And this is two generations where the two-signal chain ran five. The transferable part of the method is the doubled reference arm: running generation zero both with and without an oracle for the thing under suspicion cost one extra session per lineage and turned “it failed” into “the index is the limit, and here is its lever”.

8. Cost, and switching the creature off between words

The project set out to build a small creature trainable for one specific purpose and running on low power.

The first half had been demonstrated piece by piece. **The second half had never been measured at all** — the existing footprint report counts bytes and says in terms that the project has no memory budget, and no run anywhere reported an operation count or a spike rate. So the project could not state whether it met its own goal, in either direction.

Read the four results below together, because the third and the fourth point opposite ways. The cost is measured and is dominated by neuron count rather than synapse count; the condition index survives being frozen between words and the freeze is exact, so that result is scale-free in the gap; a single-target lesson survives too. *But what a four-word index BUYS falls by 81% of the effect*, at 3.3 standard errors — though §8.4 shows that fall is driven by the control arm rising rather than by indexed naming dropping, so it is not a demonstration that conditionality is what breaks. This section establishes that gating is free for perception and for simple teaching, and that what an index buys shrinks; it does **not** establish that the creature can be run at a few percent duty while doing its job, nor that it cannot.

8.1 The cost is bounded by the neuron count, not the synapse count

The currency is a *synapse-visit* or a *neuron update*: one of each counted once, so nothing has to be assumed about hardware. Three terms, each labelled by what it scales with, and each located by reading the enclosing function rather than by grepping for a loop.

term	**per simulated second**	**scales with**	**share**
neuron updates	1.166×10^{6}	neurons, every tick	82.6%
spike delivery	1.661×10^{5}	*event-driven*: spikes × fan-out	11.8%
plasticity sweep	8.014×10^{4}	synapses, *per reward cash-in*	5.7%
total	$\mathbf{1.412\times10^{6}}$		

The shape matters more than the total, and it is the favourable shape. The per-tick update loop contains no pass over synapses at all — every loop in it was enumerated to check — so the synaptic cost is event-driven and the one dense-in-synapses sweep is paid only when reward arrives, about twice a second. What is dense every tick is *neuron* integration, bounded by 1166 neurons rather than by 37,275 synapses: a factor of 32. On one core the creature runs 15 times faster than real time.

A correction this measurement made to itself, and it inverted the conclusion. The first version asserted that the plasticity sweep ran every tick and reported bookkeeping outweighing delivery by 226×. It does not: that sweep lives in the reward-application path, and the first run never applied reward at all, so the loop being priced executed *zero times*. The figure was not an overstatement but a count of something that did not happen. **A cost claim must name the enclosing function of every loop it prices** — "per tick" and "per cash-in" differ here by three orders of magnitude, and five of the six dense synapse loops in this kernel live in setup, sleep, pruning or instrumentation.

This is deliberately a statement about operations and sparsity and **not about watts**. The published per-operation energy figures span more than an order of magnitude and the recent ones are assumptions rather than chip measurements; they are listed in Appendix A. What they do establish is a direction: against a chip's

active power floor, which does not fall with utilisation, this creature's dynamic cost sits between 293 and 276,000 times below it across every figure available. **So operations were never the bill. Being awake is.**

8.2 The condition index survives a freeze between words, and the result is scale-free

If being awake is the bill, the lever is duty cycle, and a creature for one purpose should be off between words. Gating is modelled as a *freeze*: neither the ear nor the update step is called, so no state decays and no internal time passes, which is what a powered-down creature does. The world is still rendered every tick, so the acoustics are identical across conditions and only the creature stops.

The risk was derived from the kernel's own time constant rather than guessed. The condition index is an exponential moving average with a 1000-tick time constant, and its whole design rests on persistence. Awake, the 1100 ticks of post-reward silence decay it to $e^{-1.1}$ ≈ 33% of what the word left; frozen, it decays to 100%. So a gated creature should carry the *previous* word into the next one and smear the index.

condition	duty	first-formant bit	second-formant bit
awake	1.000	—	—
frozen after reward	0.607	−0.0143 ± 0.0172	**+0.1104 ± 0.0250**
frozen, with a longer gap	0.304	*identical to the row above, exactly*	

The prediction was refused: the first bit is untouched and the second is better — the latter replicated on an independent family of creatures (+0.154, then +0.091, both past three standard errors) before it was claimed. At two standard errors the first bit loses at most 5% of itself.

The exactness of the freeze is what converts this into a general claim. The long-gap condition returns numbers identical to the short-gap one, to every decimal, on both bits and in both families. That is forced rather than lucky: a frozen creature cannot tell how long it was off, so an 1100-tick gap and a 3900-tick gap hand it the same sequence of waking ticks, and only the denominator differs. Any difference would have meant time leaking through the gate. **So a creature that pays nothing at 30% duty pays nothing at any sparsity** — the saving is set by how often words arrive and not by anything in the creature: about 17% duty at a word every ten seconds, about 3% at a word a minute.

And the gain has a mechanism that survived its own test. The account offered was that silence is not neutral for an online clustering rule — processing it drags the stored prototypes toward a silence feature, and freezing protects them. That predicts a *monotone* dose-response in how much silence is processed, and an intermediate condition delivered one: 0.718 awake, 0.813 at 80% duty, 0.872 at 61%. It landed between rather than jumping to the gated value, which is what would have refuted the account. The intermediate step is 2.4 standard errors on its own, so what is established is the ordering; and the first formant is not monotone, though every effect on it is small on a bit already near ceiling.

8.3 A fixed-target lesson survives gating too, and the cost of the gain is a selection effect

An index is not a job. That the creature still knows which context it is in says nothing about whether reward can still *teach* it while gated, which needs the teaching loop. On two words and a fixed target, with the error reduction pooled as a ratio of sums and a jackknife over creatures:

condition	error, early	error, late	reduction	trials skipped
awake	0.4792	0.2859	40.35%	13.3%
frozen after reward	0.4412	0.2388	45.87%	9.0%

The difference is +5.53% ± 2.74%, a two-standard-error interval of +0.04% to +11.01%. **The whole interval is positive, so no cost is established** — which is what the power argument needs. The point estimate is a gain, but at two standard errors it does not clear this project's three-sigma bar and is not claimed.

A confound travels with that number and is the reason not to pursue the gain. The two conditions do not score the same trials: the gated creature skips 9.0% against the awake creature's 13.3%, and a trial is skipped when the creature said nothing. So gating also makes it *vocalise more often*, and the pooled reduction does not control for which trials enter it. The absence of a cost survives that objection; the size of the gain does not.

What would have refused the whole line: a resolved loss in either the index or the lesson, or a long-gap condition that differed from the short-gap one, which would have meant the freeze was not a freeze. And a wrong prediction worth recording: lengthening the runs was expected to shrink the lesson's error bar and it grew instead, because the jackknife is over *creatures* and the creature count had not changed. To tighten such an interval one adds units, not length.

8.4 What an index buys collapses under gating — but the control is what moves

An index is not a job and a single fixed target is not the job either. The task this creature is actually for is *four-word conditional naming*: four signals laid out as a 2×2, each to be produced when its own condition holds. Gating that is one constant, and the ungated run of the same genome already existed.

The pairing is stronger here than in §6.11 and that is what makes the test clean. Gating changes no wiring, so the same genome and the same seed give the *same creature* in both runs, differing only in whether it freezes after the reward window. Creature *r* gated and creature *r* ungated are matched units, so the comparison is properly paired — where the trim comparison could not be, because deleting modules re-rolled every later random draw.

condition	paired excess on the four-way score, n=36
awake	+0.1570 ± 0.0287 (5.5 SE)
frozen after reward	+0.0302 ± 0.0236 (1.3 SE)
paired difference	**−0.1268 ± 0.0385 (3.3 SE)**

What the index buys collapses: a resolved fall of 81% of the effect, two-standard-error interval −0.204 to −0.050, with 25 of 36 creatures lower when gated (sign test $z = +2.33$).

But that score is a difference between two arms, and splitting it changes the reading. The quantity is the indexed arm minus the no-index control, so a fall has two quite different possible causes. Paired, on the same 36 creatures:

arm	awake	frozen	paired change

indexed (host-written condition)	0.4707	0.4170	−0.0537 ± 0.0260 (2.1 SE)
no index	0.3137	0.3868	**+0.0731 ± 0.0223 (3.3 SE)**

The collapse is driven by the control rising, not by indexed naming falling. The indexed arm's own fall is 2.1 standard errors and does not clear the bar this project uses. So the stronger statement — that conditional naming does not survive gating — is *not* supported, and this document asserted it in an earlier draft of this section. What is established is narrower: **gating shrinks what an index buys, mostly by lifting the baseline that the index is measured against.**

It also refutes the mechanism that was offered with it. The carry-over account — freezing leaves the previous condition in the feature, so reward writes into the wrong row — cannot reach the arm that matters, because that arm's condition is *written by the host every trial* and carries nothing over. The indexed arm barely moved, consistent with that. The hypothesis is unsupported.

The rise in the control is left open rather than explained. Its awake value of 0.3137 sits close to this project's own measured empty-voice level for a four-way nearest-target score, which is why that statistic always carries a control at all; frozen it reads 0.3868, well above it. So gating appears to change the *output distribution* in a way that improves raw target-matching with no conditioning involved — a narrower or better-centred voice would do it. The spread columns needed to check that are already in both logs.

The retraction matters more than the result. §8.2 and §8.3 read, on their own, as licensing a creature that runs at a few percent duty while doing its job. *Nothing here supports that*, and nothing here establishes the opposite either. Gating is free for **perception** and free for a **single-target lesson**; what an index *buys* shrinks, by a route that is at least partly a shift in the baseline rather than a loss of conditionality.

This run's alternative outcome was written down in advance as "the cost localises to conditionality". The decomposition does not support it. **Recording the alternative beforehand is what made that mismatch visible rather than invisible** — it does not make the alternative correct once the data arrive.

What this does not license: abandoning duty cycling. Nothing here tests a creature gated *after* training rather than during it, which is the deployment case that matters most and is a different experiment.

9. Negative results that closed lines

Listed because the project's cost is dominated by them and because each removed a plausible mechanism from consideration.

Mechanism	Outcome
Oscillations as a rate-to-phase code	Built; spike-timing plasticity could not read it
Hippocampal pattern separation	Does not separate
Exuberant synaptogenesis	Costs 0.229; competition a further 0.051. Weight is not information

Dendritic error microcircuit	Settles and cancels nothing
Plateau-gated plasticity	Attenuates instead of selecting
Two-stage bandpass	Never appears on three seeds
Topographic tract	Oracle predicted 471 Hz; measured 1.00×
Region-shaped reward targets	A satisfied lesson still does not protect the first
Two-pool differential readout	Premise fails: only $\rho = 0.101$ of variance is shared
Adaptation current as a clock	Peak frequency follows its time constant at exponent 0.10 where 1.0 is required
Prototype freezing	Refused; it is the stability–plasticity dilemma
Replay / interleaving as shipped	Switching it off costs 0.3 SE
Reward that sees target distance	Worse, by 9.4 SE
Protocol axes (duration, blocking)	28× longer is no help; blocked teaching is harmful
Astrocytic activity gate on plasticity	Cuts learning 81% and forgetting 19%, on a monotonic dose–response
Consolidation brake on high-traffic edges	Binds hard (0.671 of the unbraked rate) and changes neither retention nor learning
Removing overall drive from the index feature	Pays +0.064/+0.161 only with *random distinct* prototype seeds, which the creature cannot produce; 1.4 SE at its own initialisation
Adaptive-resonance vigilance as the seeding rule	Recovers at most 1.6 SE of that, and at the shipped threshold commits exactly *one* slot — so the milestone built on it was measuring a constant index
An operating point where gesture and lesson compose	Twelve cells over the whole (gain, drive) plane, both axes and both corners. At the gesture bar the lesson is 3.7 SE short; the corner is strictly dominated
Per-neuron spread of the firing threshold	Costs 61% of the lesson, monotone in sigma, 3.5 SE. The threshold is already learned as the bias, so an imposed spread is noise in the quantity the rule is setting
Per-neuron spread of the membrane leak	Nothing either way, learned or imposed; bound 31%

Per-neuron spread of the refractory period	Structurally inert: the creature runs 46–81× below the ceiling the constant sets, so it governs about 1.5% of each neuron's input. −1.6% per 0.1 sigma against the threshold's −20%, and a spread holds the operating point to sigma 0.60 where the threshold broke it at 0.04
Pooling the rate regulator across an articulator group	Improves the transfer (0.34 → 0.51) and halves the learned bias, and the two offset: −0.90 SE. Every neuron receives the slice mean, so the common threshold is pushed back 3× harder and the gradient is noisier to estimate in
Exempting the readout's own coordinate from the regulator	One direction of fourteen rather than thirteen. The mechanism prediction was confirmed and the outcome did not follow: threshold push 1.02× against pooling's 3.3×, and the learned bias 0.957× with its own control falling too. The regulator's coordinate system is not what binds

10. Provenance

Every number above comes from a run of the project's own experiment binary, gated before execution on `--experiment verify` reporting PASS with the pinned hash `ad96f882becbee92` and a passing pinned-hash check. Runs that failed the gate are discarded rather than reported.

Sample sizes are creatures, not trials: a creature is one genome with one seed, grown and taught from birth. Where a paired statistic is quoted, creatures are matched by seed across arms, which is licensed by the per-job seed being a function of the repetition index alone and independent of the arm — verified by a shared baseline arm reproducing bit-identically across three separate multi-hour runs, and by a direct cross-run check of that arm against itself returning +0.0000 ± 0.0000 over 32 creatures.

Retention figures are a ratio of sums with a leave-one-out jackknife standard error, swept over a floor on per-creature learning, and are reported only where the effect holds at every floor retaining at least three creatures in both arms. Section 2.8 explains why the more obvious estimator is not used.

Run logs, genome files and the experiment source are in the project repository. The findings here are recorded at greater length, with the reasoning that produced them and the criteria that would refuse them, in the project's `README.md` and its memory store.

Appendix A. Excluded by the criterion

These project results are omitted from the body because a published result says or frames them first. They are listed so that the inclusion criterion of §1 can be checked.

The line for §8, drawn explicitly because that section is new. *Duty cycling* — powering a device down between events and waking it on a trigger — is ordinary engineering practice and is excluded; so is the observation that event-driven spiking hardware is cheap when activity is sparse. The per-operation energy figures quoted as context are other people's and are excluded: a 2026 framework paper *assumes* 1–5 pJ per synaptic operation and cites 250–500 mW of active power for a current research chip, while the only directly measured per-synapse chip figures available are older and span from 26 pJ to 941 pJ. What §8 claims is narrower and is included: that *this* creature costs 1.412×10^6 operations per simulated second with no dense-in-synapses term in its per-tick path; that freezing it between words leaves its condition index intact and its second bit better, with a monotone dose-response in silence processed; that the freeze is exact and the result therefore scale-free in the gap; and

that the lesson survives with no cost established. None of those is a statement about hardware, and §8 makes no claim in watts.

One entry needs its line drawn explicitly. The idea that behaviour depends on some directions in a population code and not others is Kaufman and colleagues', and it is excluded. What §4.2 claims is narrower and is included: in this system the map from population activity to behaviour is *analytically known rather than fitted*, so the potent direction can be differentiated instead of estimated, it is state-dependent, and its sign — not its magnitude — is what the measurements require.

The same line, drawn once more for §2.13. The scoring rule is Kuhn and Munkres's assignment problem and is excluded; that a uniform cluster prior hurts on imbalanced data is Assran and colleagues' and is excluded. What is included is the consequence of fixing that score's margins — the ceiling, its exactness, and the fact that it splits a result this project had published as one effect into two with different mechanisms. The bound is elementary once stated; nobody here had stated it, and the cost of not stating it was a candidate priced against the wrong arm by a factor of twenty-five.

Project result	**External antecedent**
Interference as a failure of contextual inference; a per-neuron mask should therefore help	Heald, Lengyel & Wolpert, COIN
Gating helps and is insufficient alone; stabilisation is the missing half	Masse, Grant & Freedman (2018)
Importance-weighted protection of earlier weights as the remedy	Kirkpatrick et al. (2017); Zenke, Poole & Ganguli (2017)
Node perturbation as the motor learning rule	Fiete, Fee & Seung
Sequence generation needs competition, not spike timing alone	Fiete, Senn, Wang & Hahnloser (2010)
Prototype freezing as a dilemma rather than a bug	Grossberg
A uniform cluster prior implicit in a competitive learning rule	MacQueen; Assran et al. (2023)
Ratio-of-sums estimation and the jackknife	Cochran; Quenouille; Tukey
Per-context reward prediction error having nothing to remove	Gadagkar et al. (2016)
Behaviour depending on output-potent and not output-null dimensions of a population code	Kaufman, Churchland, Ryu & Shenoy (2014)
Activity-dependent myelination as a consolidation mechanism, and its post-training window	The oligodendroglia literature, cited by title and DOI
An astrocyte-gated multi-timescale rule as a remedy for catastrophic forgetting	Astrocyte-gated multi-timescale plasticity (2026)
Contexts lying on a continuum rather than being discrete slots	Heald, Lengyel & Wolpert (2023)
Best-permutation agreement as the way to score an unsupervised partition, and its exact solution	Kuhn (1955); Munkres (1957), after Kőnig and Egerváry
A uniform cluster prior helping on balanced data and hurting on imbalanced data	Assran et al. (2023)

Fast associative plasticity and slow homeostatic feedback being unable to stabilise each other, and factorising the two onto separate variables as the remedy	Toyoizumi, Kaneko, Stryker & Miller (2014); Zenke & Gerstner
Intrinsic plasticity as adaptation of a neuron’s own output distribution, derived by information maximisation	Triesch (2005); Savin, Joshi & Triesch (2010)
The online prototype update whose rate is 1/wins, and the win-balance penalty beside it	MacQueen (1967); DeSieno (1988)